METRIC

MIL-DTL-17341D(SH)
13 August 2025
SUPERSEDING
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9 April 1970
(see 6.7)

DETAIL SPECIFICATION

ENGINES, GAS TURBINE, MARINE, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers marine gas turbine engines intended for use as prime movers for propulsion (mechanical drive or generator drive), electric power generation, and auxiliary drive for naval shipboard service and the enclosures and accessories to these gas turbines.

1.2 Classification. Gas turbine engines are of the following types, as specified (see 6.2).

a. Type A – Gas turbine engines intended for use in low-usage applications with annual operating hours of less than 500, such as boats, craft, hydrofoils, and boost power propulsion and for emergency generator drives, with starts typically once every 2 operating hours.

b. Type B – Gas turbine engines intended for use in high-usage applications with annual operating hours of greater than or equal to 500, such as main ship propulsion and for ship service generator drives, with starts typically once every 5 operating hours.

c. Auxiliary power units (APU) – APUs are auxiliary power generation units used to supply temporary electrical, pneumatic, or hydraulic power. They are not used for propulsion or continuous power generation and are not covered by this specification. See MIL-P-85573 for further detail on APUs.

1.3 Rating. Rating for either a type A or type B gas turbine engine is established by successfully completing the testing specified herein and for which a rating verification letter is issued by the Technical Authority (see 6.5). Each gas turbine engine that has successfully completed testing herein as either a mechanical drive or generator drive application and has received a rating by the Technical Authority (see 6.5) may be used in either mechanical drive or generator drive applications with the following exception: existing type A marine gas turbine engines previously accepted as low-usage under American Bureau of Shipping (ABS) Naval Vessel Rules (NVR) or similar previous U.S. naval ship specification requirements; such engines being offered for high-usage in a type B mechanical drive or generator application are subject to the requirements specified herein.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-907 - Antiseize Thread Compound, High Temperature

MIL-DTL-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys

MIL-DTL-5624 - Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)

MIL-PRF-8625 - Anodic Coatings for Aluminum and Aluminum Alloy

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-DTL-15730 - Coolers, Fluid, Naval Shipboard – Hydrocarbon Base Oil and Freshwater Services

MIL-DTL-16884 - Fuel, Naval Distillate

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-DTL-17813 - Expansion Joints, Pipe, Metallic Bellows, General Specification for

MIL-DTL-18866 - Fittings, Hydraulic Tube, Flared, 37 Degree and Flareless, General Specification for

MIL-G-22077 - Generator Sets, Gas Turbine, Direct and Alternating Current Naval Shipboard Use

MIL-PRF-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167

MIL-E-24572 - Extinguisher, Fire, Bromotrifluoromethane (Halon 1301) System Components (Fixed Pipe, Pneumatically Actuated, Naval Shipboard Use)

MIL-DTL-25579 - Hose Assembly, Polytetrafluoroethylene, High Temperature, Medium Pressure, General Specification for

MIL-DTL-32613 - Controller, Auxiliary-System, Naval Shipboard Use

MIL-PRF-32407 - Mounts, Resilient (Surface Ship Application)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment

MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

MIL-STD-882 - System Safety

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MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety

MIL-STD-1399-300-1 - Low Voltage Electric Power, Alternating Current

MIL-STD-1472 - Human Engineering

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

MIL-HDBK-472 - Maintainability Prediction

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

14 CFR 33 - Airworthiness Standards: Aircraft Engines

(Copies of this document are available online at www.ecfr.gov.)

NAVAL AIR SYSTEMS COMMAND (NAVAIR) PUBLICATIONS

NAVAIR 17-15-50.1 - Joint Oil Analysis Program Manual, Volume I, Introduction, Theory, Benefits, Customer Sampling Procedures, Programs and Reports

NAVAIR 17-15-50.2 - Joint Oil Analysis Program Manual, Volume II, Spectrometric and Physical Test Laboratory Operating Requirements and Procedures

NAVAIR 17-15-50.3 - Joint Oil Analysis Program Manual, Volume III, Laboratory Analytical Methodology and Equipment Criteria (Aeronautical)

NAVAIR 17-15-50.4 - Joint Oil Analysis Program Manual, Volume IV, Laboratory Analytical Methodology and Equipment Criteria (Non-Aeronautical)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9073-A2-HBK-010 - Resilient Mount Handbook, Installation and Inspection Information

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-HC-STM-010/234 - Naval Ships' Technical Manual Chapter 234, Marine Gas Turbines

S9407-AB-HBK-010 - Handbook of Shipboard Electromagnetic Shielding Practices

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN GEAR MANUFACTURERS ASSOCIATION

ANSI/AGMA 2001 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth

(Copies of this document are available online at www.agma.org.)

AMERICAN PETROLEUM INSTITUTE (API)

API RP 684 - API Standard Paragraphs Rotordynamic Tutorial: Lateral Critical Speeds, Unbalance Response, Stability, Train Torsionals, and Rotor Balancing

API STD 616 - Gas Turbines for the Petroleum, Chemical and Gas Industry Services

(Copies of these documents are available online at www.api.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1 - Power Piping

ASME BPVC - Boiler and Pressure Vessel Code

ASME MFC-26 - Measurement of Gas Flow by Bellmouth Inlet Flowmeters

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL (ASTM)

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM D189 - Standard Test Method for Conradson Carbon Residue of Petroleum Products

ASTM D240 - Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D482 - Standard Test Method for Ash from Petroleum Products

ASTM D524 - Standard Test Method for Ramsbottom Carbon Residue of Petroleum Products

ASTM D975 - Standard Specification for Diesel Fuel

ASTM D976 - Standard Test Method for Calculated Cetane Index of Distillate Fuels

ASTM D1552 - Standard Test Method for Sulfur in Petroleum Products by High Temperature Combustion and Infrared (IR) Detection or Thermal Conductivity Detection (TCD)

ASTM D2622 - Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-Ray Fluorescence Spectrometry

ASTM D2709 - Standard Test Method for Water and Sediment in Middle Distillate Fuels by Centrifuge

ASTM D2880 - Standard Specification for Gas Turbine Fuel Oils

ASTM D4294 - Standard Test Method for Sulfur in Petroleum and Petroleum Products by Energy Dispersive X-Ray Fluorescence Spectrometry

ASTM D4530 - Standard Test Method for Determination of Carbon Residue (Micro Method)

ASTM D6064 - Standard Specification for HFC-227ea, 1,1,1,2,3,3,3-Heptafluoropropane (CF3CHFCF3)

ASTM D6079 - Standard Test Method for Evaluating Lubricity of Diesel Fuels by the High-Frequency Reciprocating Rig (HFRR)

ASTM D6217 - Standard Test Method for Particulate Contamination in Middle Distillate Fuels by Laboratory Filtration

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ASTM D6304 - Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

ASTM E8/E8M - Standard Test Methods for Tension Testing of Metallic Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2314 - Gas Turbines – Acceptance Tests

ISO 3104 - Petroleum Products – Transparent and Opaque Liquids – Determination of Kinematic Viscosity and Calculation of Dynamic Viscosity

ISO 3733 - Petroleum Products and Bituminous Materials – Determination of Water – Distillation Method

ISO 3977 - Gas Turbines – Procurement

ISO 4261 - Petroleum Products – Fuels (Class F) – Specifications of Gas Turbine Fuels for Industrial and Marine Applications

ISO 4264 - Petroleum Products – Calculation of Cetane Index of Middle-Distillate Fuels by the Four Variable Equation

ISO 5165 - Petroleum Products – Determination of the Ignition Quality of Diesel Fuels – Cetane Engine Method

ISO 6245 - Petroleum Products – Determination of Ash

ISO 8217 - Products from Petroleum, Synthetic and Renewable Sources – Fuels (Class F) – Specifications of Marine Fuels

ISO 8754 - Petroleum Products – Determination of Sulfur Content – Energy-Dispersive X-Ray Fluorescence Spectrometry

ISO 10370 - Petroleum Products – Determination of Carbon Residue – Micro method

ISO 10380 - Pipework – Corrugated Metal Hoses and Hose Assemblies

ISO 11042 - Gas Turbines – Exhaust Gas Emission

ISO 12156-1 - Diesel Fuel – Assessment of Lubricity Using the High-Frequency Reciprocating Rig (HFRR) – Part 1: Test Method

ISO/IEC/IEEE 12207 - Systems and Software Engineering – Software Life Cycle Processes

ISO 14596 - Petroleum Products – Determination of Sulfur Content – Wavelength-Dispersive X-Ray Fluorescence Spectrometry

ISO/IEC-IEEE 15288 - Systems and Software Engineering – System Life Cycle Processes

ISO 17359 - Condition Monitoring and Diagnostics of Machines – General Guidelines

ISO 19860 - Gas Turbines – Data Acquisition and Trend Monitoring System Requirements for Gas Turbine Installations

ISO 20816-3 - Mechanical Vibration – Measurement and Evaluation of Machine Vibration – Part 3: Industrial Machinery with a Power Rating Above 15 kW and Operating Speeds Between 120 r/min and 30 000 r/min

ISO 21940-11 - Mechanical Vibration – Rotor Balancing – Part 11: Procedures and Tolerances for Rotors with Rigid Behaviour

ISO 21940-12 - Mechanical Vibration – Rotor Balancing – Part 12: Procedures and Tolerances for Rotors with Flexible Behaviour

(Copies of these documents are available online at www.iso.org.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 72 - National Fire Alarm and Signaling Code

(Copies of this document are available online at www.nfpa.org.)

SAE INTERNATIONAL

SAE AIR4391 - Industrial and Marine Gas Turbine Engine Starting Systems

SAE ARP1420 - Gas Turbine Engine Inlet Flow Distortion Guidelines

SAE ARP1587 - Aircraft Gas Turbine Engine Health Management System Guide

SAE AS150 - Hose Assembly, Type Classifications of, Basic Performance and Fire Resistance

SAE AS755 - Aircraft Propulsion System Performance Station Designation

SAE AS1072 - Sleeve, Hose Assembly, Fire Protection

SAE AS1424 - Hose Assemblies, Metal, Medium Pressure, High Temperature

SAE/ANSI B92.1 - Involute Splines and Inspection

SAE J1775 - Bleed-Air Pneumatic Systems for Gas Turbine Equipped Marine and Amphibious Craft

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 864 - Standard for Control Units and Accessories for Fire Alarm Systems

(Copies of this document are available online at www.ul.com/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the proposed gas turbine engine shall be subjected to first article inspection in accordance with 4.2. This shall include a 3000-hour endurance test in accordance with 4.6.4 unless a design assessment process (DAP) in accordance with appendix A indicates a lower number of endurance test hours should be considered for approval by the Technical Authority (see 6.5). The gas turbine engine shall be identified with a model and a proposed rating (see 3.1.1 and 6.2). The gas turbine engine model shall be identified for usage as one of the two gas turbine engine types (see 1.2) in a Gas Turbine Qualification Letter. Gas turbines may be procured as engines alone, engine sections, engines and accessories, engine control systems, or complete engine packages and enclosures (see 6.2).

3.1.1 Rated power. A Navy rating shall be identified for the gas turbine engine (see 6.2). The rating of the gas turbine engine shall be a flat rating under Navy Standard Day (NSD, see 3.3.1) conditions. An NSD flat rating results in a constant power output at ambient temperatures up to 38 °C (100 °F). A flat rating curve is illustrated in figure 1. Industry typically uses an ISO 3977 tent curve rating whereby the power output of the gas turbine engine varies with the ambient air temperature. The U.S. Navy does not use the ISO 3977 tent curve rating approach. Uprating of existing gas turbine designs shall be in accordance with appendix A.

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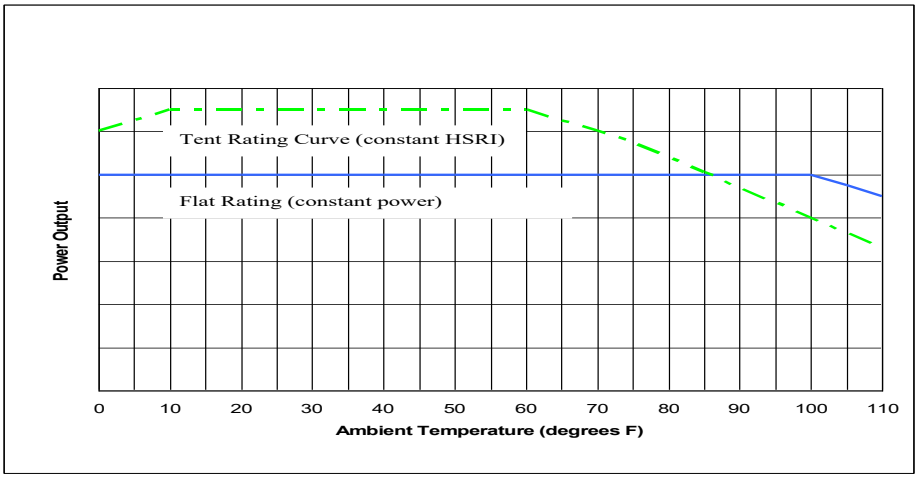


FIGURE 1. Gas turbine rating curves.

3.1.1.1 Power range. The gas turbine engine power capability determined for ambient temperatures from -29 °C (-20 °F) to +38 °C (+100 °F) shall be rated power.

3.1.2 Specific fuel consumption. The specific fuel consumption of the proposed gas turbine engine shall be identified across the gas turbine engine power band for the NSD condition (see 6.2).

3.1.3 Operability. The gas turbine engine shall be designed for continuous operation at each point of the specified speed range and power range up to and including rated power. The gas turbine engine manufacturer shall define the period between hot section (combustor and turbine) repairs when operating continuously at rated power. Other requirements for operability are as follows:

a. The gas turbine engine design shall accommodate transient thermal gradients following emergency shutdowns if the engine is immediately restarted.

b. The gas turbine engine shall be designed to allow for operation without damage up to the overspeed trip speed of the gas turbine engine.

c. The gas turbine engine shall be designed for extended shutdown periods of up to 30 days without requiring any special maintenance procedures.

3.2 Materials. Materials used to construct the gas turbine engine, its accessories, and enclosures (when required [see 6.2]) shall be in accordance with API STD 616 for industrial gas turbine engine designs and aero derivative standards, such as 14 CFR 33 subpart B, for aero derivative gas turbine engines (a gas turbine engine derived from an aircraft application) except as modified to meet the requirements of material for marine operation and for structural members subjected to the shock test requirements. Materials with elongations of less than 10 percent in accordance with ASTM E8/E8M or ASTM A370 shall not be used for structural members subjected to the shock requirements. Other general requirements for materials are as follows:

a. All materials shall be resistant to or protected to resist salt water corrosion and corrosion from other fluids used in operating, maintaining, or servicing the gas turbine engine in a marine environment for the temperatures encountered for the projected powers in meeting the life requirements of 3.4.1.

b. Only corrosion-resistant materials shall be used in seawater system piping and components.

c. Any material used for the vibration or shock isolation mounting system shall comply with the in-service environmental requirements of MIL-PRF-32407.

d. Contact between dissimilar metals shall be avoided. An interposing material minimizing or eliminating galvanic effect shall be used.

e. Aluminum parts, other than blowers and fuel pumps, shall be electrically isolated from dissimilar metals and protected from corrosion.

f. Copper-bearing alloys and aluminum shall not be combined in the cooling water system.

g. The cooling inhibitor shall be compatible with the alloys in contact with the water.

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h. Copper shall not be used in fuel or lubricating oil piping, except for Monel-type (Unified Numbering system [UNS] 04400) copper alloys or equal, which may be used in lubrication oil piping.

3.2.1 Untreated surfaces. Unpainted aluminum or aluminum alloy surfaces shall be treated to conform to MIL-PRF-8625. Surfaces to be painted shall be treated to conform to MIL-DTL-5541. Internal surfaces and surfaces in contact with oil shall not be painted. Untreated steel alloy shall be treated to preclude or control corrosion.

3.2.2 Threaded surfaces. When threaded surfaces or faying surfaces require disassembly in service, an anti-seize compound shall be used within the limitations of MIL-PRF-907. Zinc-plated steel through bolting and nuts shall be used to assemble alloy parts, particularly in highly stressed assemblies or those subject to vibrating loads. When tap bolts or machine screws must be used, these shall be threaded into passivated steel inserts in the aluminum alloy. The steel inserts shall be secured in such a way as to prevent them from turning or backing out. No alloys of copper, such as bronze or brass, shall be used in threaded contact with aluminum alloys. All threads shall be in accordance with FED-STD-H28.

3.2.3 Welding. Welding for the gas turbine and piping within the gas turbine enclosure (if applicable), shall be in accordance with API STD 616. Welding of aero derivative gas turbines shall be in accordance with the aero standards of 14 CFR 33 or other Technical Authority-approved welding standards. Piping interfaces with the ship piping systems shall be in accordance with S9074-AR-GIB-010/278.

3.2.4 Prohibited materials. The following materials are prohibited on or in the engine, base, enclosure, sensors, actuators, or other gas turbine-related hardware, with exceptions as allowed by T9070-AL-DPC-020/077-2:

a. Asbestos.
b. Mercury, mercury compounds, or instruments containing mercury or mercury compounds.
c. Magnesium alloys.
d. Paint, plastic, zinc chromate or zinc coating on the interior of gear housings, lubricating-oil sumps, or any other interior surfaces that will be in contact with lubricating oil, fuel oil, or hydraulic oil.
e. Cadmium (e.g., cadmium plating shall not be used on engine parts including fasteners).
f. Lead.
g. Beryllium (if there is possibility of human exposure during maintenance).
h. Chrome plating.
i. Hexavalent chromium.
j. Radioactive material.

3.2.5 Health hazard. Materials shall be in accordance with MIL-STD-882.

3.2.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental conditions.

3.3.1 NSD. NSD reference conditions for marine gas turbine engine rating shall be in accordance with table I.

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TABLE I. Standard rating condition for gas turbine engines.

Parameter	Surface ship standard rating conditions
	NSD
Inlet air temperature 1/	38 °C (100 °F) 2/
Inlet barometer air pressure dry 1/	101.325 kilopascals (kPa) (1.01325 Bar, 29.92 inches of Hg) 2/
Seawater cooling inlet temperature	35 °C (95 °F)
Relative humidity	60 %
Inlet air system depression after filter 1/	1.0 kPa (0.01 Bar, 4 inches of H2O) – total
Exhaust system back pressure	1.5 kPa (0.015 Bar, 6 inches of H2O) – static
Fuel lower heating value	42,798 kJ/kg (18,400 Btu/lb)

NOTES:

1/ Measurement locations for these specified conditions shall be in accordance with ISO 2314.

2/ These standard condition values shall be used within the ISO 2314 power correction when this correction is applied.

3.3.2 Salt air environment. The gas turbine engine will be operated in a marine environment, which includes a level of salt ingestion. The gas turbine engine shall be tested for salt ingestion in accordance with 4.6.2.2.

3.4 Gas turbine engine design.

3.4.1 Gas turbine engine design life and reliability. Design life and reliability shall be as required in 3.4.1.1 through 3.4.1.3 and 3.4.2.

3.4.1.1 Design life. The gas turbine engine shall be designed and constructed for a minimum service life of 30 years. Type A and type B gas turbine engines are typically used for main propulsion or electrical power generation. The operating duty profiles in starts and hours shall be as follows:

a. Type A engines. Assuming an annual usage of 300 hours/year, the type A gas turbine engine design operating life shall be a minimum of 9,000 hours.

b. Type B engines. Assuming an annual usage of 2000 hours/year, the type B gas turbine engine design operating life shall be a minimum of 60,000 hours.

3.4.1.2 Mean installed engine life (MIEL). The MIEL (see 6.6.6) of a type A gas turbine engine shall be a minimum MIEL of 5,000 hours. The MIEL of a type B gas turbine engine shall be a minimum MIEL of 12,500 hours when based on 525 starts and a power duty cycle as specified in table II at ambient air temperature of 24 °C (75 °F), burning NATO F-76 (MIL-DTL-16884) fuel, not exceeding 0.4 weight percent of sulfur, ingesting 0.005 parts per million (ppm) of salt in the inlet air to the compressor, and without exceeding a maximum performance degradation of 2 percent.

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TABLE II. MIEL-calculated power distribution for type B gas turbine engines.

Power condition (%)	Time (hours)
100	1,250
90	1,250
80	1,250
70	1,750
50 or below	7,000

3.4.1.3 Mean time between failure (MTBF). Unless otherwise specified (see 6.2), type A gas turbine engines shall have a have a minimum MTBF of 1,500 hours based on 500 operating hours per annum. Type B gas turbine engines shall have a minimum MTBF of 2,000 hours in mechanical drive applications and 3,000 hours in electrical generator applications based on 500 starts per annum.

3.4.2 Hot section repair interval (HSRI). The gas turbine "time at rated power" (TARP) shall result in a gas turbine HSRI (see 6.6.4) of a minimum of 1,250 hours for type A engines when operating at rated power and speed on an NSD. The gas turbine TARP for type B engines shall result in an HSRI of 2,500 hours for type B engines of less than or equal to 12,000 kilowatts (kW) (16,086 BHP) and an HSRI of 2000 hours for engines of greater than 12,000 kW (16,086 BHP) when operating at rated power and speed on an NSD.

3.4.3 Construction. Gas turbine engine designs shall be in accordance with API STD 616 and shall be designed to operate in a marine, shipboard environment that includes ship roll, yaw, and pitch. The gas turbine engine shall be designed to contain any failure of blades or blade attachment devices.

3.4.3.1 Compressor(s). The compressor materials and coatings shall be selected to inhibit marine operation-induced corrosion and deterioration and comply with 3.2.

3.4.3.2 Turbine(s). The turbine materials and coatings shall be selected to inhibit marine operation-induced corrosion and deterioration throughout the proposed turbine operating temperature range and comply with 3.2.

3.4.3.3 Combustor(s). The combustion system shall be designed to operate without erosion, plugging, carbonization, cracking, or other failure for the duration of the engine hot section repair interval. The combustion system shall provide proper flame front location and length, combustor residence time, equivalence ratio, and combustor outlet temperature distribution and flame stability. The combustion system shall also provide low smoke, low carbon formation, combustor durability, and reliable starting and operation. For a can-annular type combustor design, wherein there are a series of individual combustors around the 360-degree circumference of the engine, the individual combustor cans shall be replaceable without engine removal and with minimal disassembly of other engine components.

3.4.3.4 Power turbine. During overspeed conditions, the power turbine shall not suffer damage at rotational speeds up to 110 percent rated revolutions per minute (rpm). The power turbine shall meet the requirements for idle and drop load operation. When the gas turbine engine design uses a free power turbine, the free power turbine shall be designed to allow indefinite operation of the gas turbine at idle with the ship's power turbine brake engaged in a ship mechanical drive application.

3.4.3.5 Rotors and discs. The rotors and discs shall be designed with sufficient strength to withstand the abnormal conditions listed below without exceeding the manufacturer's allowable dimensional limits and without any evidence of incipient cracking or distortion.

a. All rotors and rotating disc components shall be designed to withstand speeds of 115 percent of maximum allowable operating speed at maximum allowable gas temperature for a period of 5 minutes.

b. All rotors and rotating disc components shall be designed to withstand gas temperatures at least 42 °C (75 °F) in excess of the maximum allowable gas temperature and at maximum allowable operating speed for a period of 5 minutes.

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c. All rotors shall be designed to be capable of withstanding momentary speed excursions up to 110 percent of the trip speed settings at maximum allowable operating temperature without physical damage to the rotor or gas turbine.

d. All rotors and rotating disc components shall be designed to have a minimum calculated burst speed of not less than 122 percent of the maximum allowable speed with a bore metal temperature at maximum allowable design metal temperature.

e. Rotors shall be designed to accept the maximum steady-state and transient torque levels required or imposed by any anticipated operating condition within the complete operating capability range of the gas turbine. Where loads such as the following exceed the operating range of the gas turbine, a shear type coupling shall be installed:

(1) Overload torque resulting from generator out of phase synchronization, where applicable.

(2) Deceleration torque resulting from rapid stopping or seizure of the driven equipment.

(3) Excessive pulse load application and removal.

f. Turbine rotors shall be capable of safe operation in the event of sudden loss of 100 percent load. Blades, discs, or shafts shall not fracture or separate as a consequence of the resulting overspeed condition.

g. Rotor shaft journal areas shall not be chrome-plated.

h. Each rotor and disc shall be clearly marked with a unique identification number. This number shall be in an accessible area that is not prone to damage.

3.4.3.6 Blades. The tips of rotating blades and labyrinths of shrouded rotating blades shall be designed to allow the unit to start up under ambient conditions of (-29 to +49 °C [-20 to +120 °F]) following a normal shutdown. During an emergency shutdown without the normal cool down period, the gas turbine shall be able to restart within the limitations as defined by the gas turbine engine manufacturer. Blade and vane clearances shall be designed to prevent damage to the rotors or stators during emergency shutdowns.

3.4.3.6.1 Rubbing. When the design permits rubbing during normal startup, the component shall be designed to be rub-tolerant. Rubbing shall not prevent the engine from meeting specified performance.

3.4.3.6.2 Natural blade frequencies. Natural blade frequencies shall not coincide with any source of excitation sufficient to contribute to high cycle fatigue failures.

3.4.3.7 Bearings. Bearing design shall be based on load-life analysis, which, at a minimum, shall consider the following:

a. Rotor weight reaction.
b. Vibratory loading.
c. Preloading.
d. Misalignment.
e. Gear loads.
f. Combined thrust and radial loads under continuous power output variation.
g. Off-design point loads.
h. Blade-out loads (loss of blades).
i. Surge loading.
j. Ship motions.
k. Shock.

3.4.3.7.1 Rolling element bearings. Rolling element bearing design shall be based on the requirements of API STD 616, except rating and lifing shall be based on NSD rating conditions rather than ISO standard day rating conditions.

3.4.3.7.2 Hydrodynamic radial bearings. Hydrodynamic radial bearing design shall be based on the requirements of API STD 616, except rating and lifing shall be based on NSD rating conditions rather than ISO standard day rating conditions.

3.4.3.7.3 Hydrodynamic thrust bearings. Hydrodynamic thrust bearing design shall be based on the requirements of API STD 616, except rating and lifing shall be based on NSD rating conditions rather than ISO standard day rating conditions.

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3.4.3.8 Internal shafting. Internal gas turbine engine shafts shall be designed in accordance with API STD 616.

3.4.3.9 Casings. Gas turbine engine casing design shall be in accordance with API STD 616.

3.4.3.10 Auxiliary gearbox. Auxiliary gears shall be rated for at least 110 percent of the power developed by any starting and helper drivers coupled to the gears and of the power required by any driven auxiliaries. No part of the accessory gearbox shall prevent individual removal of any one external component or accessory mounted on the gearbox.

3.4.3.10.1 Auxiliary gearbox mounting. The gearbox shall be mounted such that there is no amplification of vibratory input into accessories mounted on the gearbox. Accessories mounted on the gearbox shall be capable of withstanding amplification of engine vibratory input induced by the gearbox.

3.4.3.10.2 Gearbox durability and strength. Gears shall be designed in accordance with ANSI/AGMA 2001.

3.4.3.10.3 Gear torque. Gears and shafting (including couplings) for generator sets shall be able to withstand the torque resulting from a short circuit across the generator terminals.

3.4.3.11 Splines. All drive splines shall be positively lubricated during operation. Design of splines for main engine shaft couplings and accessory drives shall conform to applicable standards, such as SAE/ANSI B92.1 or other equivalent recognized standards. Splines of appurtenances, such as starters that are only used temporarily and are thereafter non-rotating when the engine is in use, are exempt from these requirements.

3.4.3.12 Couplings. Requirements for couplings between the gas turbine and the driven equipment shall be as follows:

a. Couplings shafts shall be sized to withstand rated torques and transient loads including electrical fault torques (ship service generator sets). For electrical generator coupling shafts, the coupling shall have a shear section.

b. Coupling shafts shall be designed to withstand the maximum allowable forces resulting from the maximum allowable deflection of the coupling shaft. The engine shall be capable of withstanding that loading.

c. The thrust force on the coupling design shall not exceed the allowable thrust force on the gas turbine.

d. For gas turbines greater than 7,460 kW (10,000 shaft horsepower [shp]), the coupling shaft shall require no lubrication.

e. The couplings and spacers shall be dynamically balanced independently to a tolerance for the maximum continuous speed of the rotating assemblies they are attached to. The coupling halves, distance piece (if applicable), and other components that make up the coupling shall be match marked. Where bolts are interchangeable, they shall be interchangeable without affecting balance.

f. The coupling shall be designed to take the necessary relative movements of the coupled equipment. These relative movements may be in an axial plane, in a radial plane, or in a combination of axial and radial planes.

g. Flexible couplings, when utilized, shall be designed for high displacement excursions due to high-impact (H.I.) shock in accordance with MIL-DTL-901.

h. Couplings shall meet the requirements of API STD 616.

i. Where applicable, couplings shall be designed to permit laser alignment.

3.4.3.13 Seals. Seals shall meet the requirements of API STD 616.

3.4.3.14 Rotor dynamics and vibration. The engine system shall be free of destructive vibration at all engine speeds and powers, including steady state and transient operation, throughout the environmental conditions and operating envelope of the engine system. The standards of ISO 21940-11 for rigid response rotors and of ISO 21940-12 for flexible rotors shall be applied.

a. The gas turbine engine shall be in accordance with the rotordynamic requirements of API STD 616 and API RP 684.

b. Natural frequencies of the rotors, bearings, casing, frames, and supports shall not exist inside the operating speed range (see 6.6.9) of the gas turbine engine from idle speed to overspeed trip speed.

c. Aeroderivative gas turbines often exhibit structural resonances within their operating speed range. If resonances are predicted, the gas turbine engine manufacturer or packager shall demonstrate a history of reliable operation of the proposed aeroderivative design.

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3.4.3.14.1 Prohibited speed ranges. There shall be no prohibited speed ranges between idle and 115 percent of engine rated speed, except for those speed ranges automatically excluded by the engine control system.

3.4.4 Ignition system for gas turbine engines. The gas turbine engine shall be equipped with an ignition system.

3.4.4.1 Exciter. There shall be a minimum of two ignition exciters unless a reliability analysis supports use of a single exciter to meet gas turbine engine reliability requirements. Exciters shall last the overhaul life of the engine.

3.4.4.2 Igniters. The igniters shall operate without failure for a minimum of 6,000 engine operating hours. Igniters shall be removable without dismantling the combustor assembly or removing the engine. There shall be a minimum of two igniters.

3.4.5 Fuel system for gas turbine engines.

3.4.5.1 Fuel nozzles and fuel manifold. Design of the fuel nozzles and the fuel manifold, as well as air assist features, if provided, shall ensure complete drainage upon shutdown to prevent internal deposits from forming due to heat soak. Fuel nozzles shall be removable without dismantling the combustor assembly or removing the engine. Fuel nozzles shall be designed to operate without erosion, plugging, or carbonization for the duration of the engine overhaul cycle. Fuel nozzles shall be interchangeable without need for field calibration.

3.4.5.2 Fuel metering valve. The fuel metering valve shall be mounted off the gas turbine engine where possible. The fuel metering valve shall provide an electronic output signal defining its position from 0 to 100 percent open.

3.4.5.3 Fuel shutoff valve. The fuel shutoff valve shall be separate from the fuel metering valve. There shall be dual redundant fuel shutoff valves mounted in series unless a reliability analysis supports Technical Authority approval of use of a single valve. One of the two shutoff valves is allowed to be embedded within the fuel metering valve. Shutoff valves shall close rapidly and completely. Automatic shutoff valves shall be closed when de-energized. Snap rings shall not be used. Shutoff valves shall be located to minimize the fuel between the shutoff valve and the engine-mounted fuel manifold.

3.4.5.4 Fuel piping and drainage. Positive means shall be provided to drain all unburned fuel from the combustor, turbine, exhaust collector, or other collection areas to the gas turbine enclosure fuel drainage system. Fuel drains and drain lines shall be plumbed to enclosure, base, and floor drains and not allowed to drain on the engine enclosure deck. Fuel piping shall be so arranged as to avoid air pockets between the engine and the base deck drains. Where feasible, the fuel drain valve should be mounted off-engine.

3.4.5.5 Fuel system materials. Metallic lines shall be corrosion resistant and comply with 3.2.

3.4.5.6 Fuel filter. The engine shall have an engine-supplied fuel filter. The filter can be either off-engine or on-engine mounted. The filter shall contain an internal relief valve and a differential pressure warning signal (for example, a switch) that shall activate at a pressure lower than would cause the relief valve to lift. The filter piping arrangement shall prevent the accumulated contaminants from entering the system through the relief valve.

3.4.5.7 Fuel purge. The gas turbine engine fuel system shall have a fuel purge capability. The fuel purge capability shall be electronically actuated and part of the control system.

3.4.5.8 Fuel. The engine shall operate using fuel in accordance with MIL-DTL-16884 (NATO F76) and be compatible with use of MIL-DTL-5624 (NATO F44), ASTM D975 grade 2-D, ASTM D2880, ISO 8217 category DMA (with a maximum sulfur limit of 1.0 and flashpoint below 88 °C [190 °F]), and ISO 4261 fuels.

3.4.5.9 Fuel supply operating parameters. The fuel system shall be designed to accommodate fuel provided by the shipboard system (at the connection to the engine) having the following parameters:

a. Fuel supply pressure: 34.5 to 345 kPa (5 to 50 pounds per square inch [lb/in^2], gauge).
b. Fuel temperature: shall be such that the fuel viscosity is within the engine manufacturer's requirements.
c. Fuel purity:
   (1) Maximum 100 parts per million (p/m) total water.
   (2) Maximum 2.5 milligrams per liter (mg/L) suspended solids after having passed through a nominal 10-micron filter.

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3.4.5.10 Fuel line protection. Fuel lines inside the engine enclosure shall be protected or routed to minimize impact turbine blade or disk failure. All fuel lines within the engine module (see 6.6.7) or engine enclosure shall be routed forward of those planes of rotation of the turbines.

3.4.6 Lubricating oil system for gas turbine engines. Each gas turbine engine shall be equipped with a dedicated lubricating oil system. The lubricating oil system shall be capable of adequately lubricating the unit throughout the operating range. Gas turbine engines shall provide or maintain oil to all bearings and maintain seal integrity whenever the shaft is rotating, including coast down, and while running on an inching motor (turning gear), if provided. Provisions shall be incorporated to prevent oil from entering the air or gas stream and causing excessive oil consumption. Bearing drains shall be of sufficient capacity to prevent oil pressures from building up excessively at the bearings and prevent sump flooding. The system shall preclude the possibility that metal particles from failure in a single area will be distributed to other areas, causing secondary damage. A pre-operation lubricating oil system shall be supplied for any engines having bearings that do not use rolling elements. No material containing more than 5 percent copper, except Monel or similar, shall be used in the cooler lubricating system. Lubricating oil supplied from the lubricating oil storage to the engine-mounted lubricating oil pump shall have a positive supply pressure.

3.4.6.1 Gear, clutch, coupling, and bearing lubrication system. Other than hydraulic start system components, accessory gears, clutches, couplings, and bearings shall use the same lubricating oil system as that used in the engine.

3.4.6.2 Lubricating oil. The gas turbine engine shall utilize and operate with non-proprietary lubricating oil in accordance with MIL-PRF-23699, or MIL-PRF-17331 (2190 TEP) in cases where hydrodynamic bearing are used, without excessive lubricating oil consumption as defined by the engine manufacturer.

3.4.6.3 Drains. The gas turbine engine oil drainage system shall be designed to prevent the accumulation of oil within the engine. All oil drainage shall be collected and plumbed to gas turbine base or floor connections and drains.

3.4.6.4 Lubricating oil tank. The lubricating oil tank fill fitting shall be located so that the tank can be filled without special tools. The fill cap shall be fastened to the unit by a chain or a similar captive method. An oil level indicator shall be provided and shall be located in an accessible position during gas turbine engine operation. Each oil tank shall have an oil level measuring and filling arrangement that allows measuring and filling while the unit is running. The lubrication system replenishment connection to the tank shall be located such that the engine lubrication system can be replenished while the engine is in operation.

3.4.6.5 Readouts. Provisions shall be made to measure oil pressure and temperature. Large engines (rated at 7,460 kW [10,000 shp] or higher) shall have, as a minimum, a combined scavenge oil temperature indication. All engines shall have an operating level mark on the oil level gauge or dipstick.

3.4.6.6 Breather system. The breather and vent system shall prevent oil in liquid form from being lost through the breather and vent when the unit is operated under conditions specified herein. All oil sumps shall be vented to an oil mist separator device, such as a precipitator, before being vented to the engine exhaust stack. The oil venting system shall include adjustable orifices to allow regulation of oil venting pressures.

3.4.6.7 Cleaning. The lubricating oil tank shall permit complete internal cleaning through access holes if the tank is built into the foundation or subbase or by removing and flushing if it is removable. Hideout points for dirt shall be minimized in the systems by having no undrainable traps and allowing only rounded, rather than sharp, corners. The engine shall have a placard warning of lubricating oil hazards if hazards exist, and this placard shall be attached adjacent to enclosure access openings when an enclosure is provided.

3.4.6.8 Filtration. Lubricating oil filters shall be duplex filters that can be switched during gas turbine engine operation. Filters shall have either an internal or external relief valve to the filter case to bypass oil around the filter and shall have a pressure switch or gauge with a warning mark to indicate to the operator when the differential pressure (inlet pressure minus outlet pressure) across the filter is approaching the filter differential pressure bypass limits. Lubricating oil filters and strainers shall be designed to prevent the accumulated contaminant from entering the system through the relief valve.

3.4.6.9 Lubricating oil cooling. A lubricating oil cooler shall be provided to maintain oil temperature within allowed limits for the selected class of lubricating oil. A gas turbine engine that is proposed for mechanical drive propulsion shall use main reduction gear lubricating oil as the coolant medium for the gas turbine lubricating oil

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system. For type A engines of less than 4500 KW (6032 shp), fuel oil cooling of lubricating oil is permissible. For gas turbine generator sets with seawater coolant, the coolant shall be at a lower pressure than the lubricating oil within the heat exchanger when operating. If freshwater is the coolant, the complete freshwater system, including expansion tank and heat exchanger for cooling the freshwater, shall be provided. The capacity of the cooler shall be adequate to maintain lubricating oil temperatures within operating limits for standard conditions at all loads up to 110 percent of rated load when operating with seawater inlet temperatures of 1 to 35 °C (35 to 95 °F). Coolers shall be in accordance with MIL-DTL-15730.

3.4.6.10 Pressure relief. Means shall be provided to limit the lubricating oil pump(s) discharge pressure to prevent over pressurizing the system.

3.4.6.11 Seawater piping. Seawater piping shall be in accordance with MIL-STD-777.

3.4.7 Bleed air system. When the gas turbine engine design utilizes multiple compressors operating at different speeds, the gas turbine design shall include a compressor interstage bleed system to prevent compressor surge. Additionally, the gas turbine engine shall be designed to provide a customer bleed air system for off-engine usage.

3.4.7.1 Customer bleed air. A bleed air system shall be provided to extract compressor discharge air from the gas turbine engine. Bleed air shall not be extracted from the exhaust piping. Bleed air shutoff valves shall be made of corrosion-resistant materials. The bleed air valves shall be quick closing. The gas turbine engine portion of the customer bleed air system shall be in accordance with SAE J1775.

3.4.7.2 Enclosure bleed air piping. Bleed air flexible piping shall be in accordance with ASME B31.1. Bleed air flexible piping shall include a corrosion-resistant metal overbraid. An internal lining shall be provided to enhance fatigue life and be capable of withstanding temperatures up to 344 °C (650 °F) without damage. Bleed air piping shall be designed to grow and contract in only the axial direction with no lateral movement. Bleed air flexible piping shall be mounted, routed, and retained to ensure that actuation, operation, and securing of the bleed air does not cause contact between the bleed air flexible pipe and other components.

3.4.7.3 Bleed air to ship interface piping. The bleed air piping interface to ship's piping, if welded, shall comply with S9074-AR-GIB-010/278 class P-1 for design pressures and temperatures exceeding 2069 kPa (300 lb/in$^{2)}$ and 344 °C (650 °F) or class P-2 for design pressures and temperatures not exceeding 2069 kPa (300 lb/in$^{2)}$ and 344 °C (650 °F).

3.4.8 Starting system. The gas turbine engine installation shall be provided with a system for starting and motoring each gas turbine engine. The gas turbine engine start system shall be in accordance with SAE AIR4391 and the requirements herein. The starting system shall be able to start the gas turbine at any ambient temperature from -29 °C (-20 °F) to +49 °C (+120 °F). The starting system shall be able to perform a start of the gas turbine engine within 180 seconds. The start system shall be capable of starting and motoring the gas turbine engine and rotating the gas turbine engine during water washing without affecting the predicted life of the starter. The start system shall be designed for starter re-engagement without the engine core speed going to zero rpm. A mechanical or control system interlock shall be provided to prevent the accidental engagement of the starter during gas turbine engine operation. Options for gas turbine engine start systems are:

a. Pneumatic.
b. Hydraulic.
c. Electric.
d. Combustion.

3.4.8.1 General starting system requirements.

3.4.8.1.1 Multiple gas turbine engines. Where multiple gas turbine engines are installed, each gas turbine engine shall have a separate and dedicated start system. Components external to the dedicated start system, such as sources of starting power (i.e., pneumatic, hydraulic, electricity, or other power sources), may be shared, but shall be designed such that the failure of an individual gas turbine engine or gas turbine engine starting equipment does not affect the ability to start any other gas turbine.

3.4.8.1.2 Duty cycle. The starter shall be capable of performing three gas turbine engine starts within eight minutes without need for starter maintenance, cooldown, or reset.

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3.4.8.1.3 Replacement. If starters cannot be manually removed and replaced by one person, provisions shall be provided to support handling, safe removal, and replacement of the starter unit.

3.4.8.2 Pneumatic starting system. If a pneumatic starting system is selected as the method for starting the gas turbine engine, it shall include the following:

a. Air starter.

b. Regulating and shutoff valves.

c. Solenoid valves. A combination pressure regulating/solenoid shut-off valve is acceptable.

3.4.8.2.1 Start air source. The start air source requirements for a pneumatic start system shall be specified (see 6.2).

3.4.8.2.2 Flexible air start hose. Enclosure flexible air lines shall be in accordance with ISO 10380, type 2, or equivalent, except that they shall be covered with one or two layers of wire braid reinforcement. Braid cover shall be firmly held in place by end fittings.

3.4.8.2.3 Start air valve. The air start valve body and flow path shall be made of corrosion-resistant steel.

3.4.8.2.4 Moisture separator. Where the ship's bleed air system is the source of air for pneumatic air starting, provisions shall be provided within the gas turbine air start system to minimize the susceptibility of the gas turbine air start system to damage from the amount of entrained water from the ship's pneumatic air system.

3.4.8.3 Hydraulic starting system. If a hydraulic start system is selected as the method for starting the gas turbine engine, each gas turbine engine shall be provided with its own hydraulic start motor and valves to provide the requisite torque for starting and motoring over the complete range of ambient conditions using hydraulic oil from the hydraulic start unit (HSU). The gas turbine engine hydraulic start system shall be designed for the full range of temperatures, pressures, and flow rates required to supply requisite torque from the HSU. The starter shall be capable of starting and motoring the gas generator and gas turbine and rotating the gas generator and gas turbine during water washing without affecting the predicted life of the starter when operated in accordance with the starter manufacturer's limitations.

3.4.8.3.1 HSU. For gas turbine engines located in different machinery spaces, each gas turbine engine machinery space shall be provided with a dedicated HSU. All HSU drain connections and drip pans shall be piped to a centralized drain for ready connection to the ship's oily waste drain system. Each HSU drip pan shall be provided with an appropriately sized coaming.

3.4.8.4 Electric starting system. If an electric starting system is selected as the method for starting the gas turbine engine, it shall include, but not be limited to, the following:

a. Engine-mounted electric starter and clutch assembly.

b. Drive and control cabinet.

c. Communication with the gas turbine engine controller.

3.4.8.5 Combustion starting system. If a combustion starting system is selected as the method for starting the gas turbine engine, it shall consist of a small gas turbine attached via a clutching mechanism to the larger main gas turbine engine to provide starting and motoring of the gas turbine engine. The combustion start gas turbine engine shall be started via an electric starter. Any battery or separate controller shall be mounted outside the gas turbine engine enclosure and have a dedicated charging system. Other requirements for combustion starting systems shall be as follows:

a. The combustion starting system shall be capable of motoring for turbine cool down, combustor fuel purge, or water washing for any number of cycles within the duty cycle for the starting system as specified by the starting air system manufacturer.

b. The combustion starting system shaft line shall be installed parallel to the gas turbine engine shaft line. Failure of the combustion starting system gas turbine shall not impact or cause failure of the gas turbine engine it is dedicated to start.

c. The heat rejection from the combustion starting system shall not impact the reliability of other electronics within the gas turbine enclosure.

d. Installation of the combustion starting system shall not impact the maintainability of the gas turbine engine.

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e. The combustion starting system lubricating oil system may be interconnected with the lubricating oil system of the engine that it is starting.

f. If the combustion starting system draws air from the gas turbine engine enclosure, the combustion starting system and the gas turbine engine it is dedicated to start shall be designed such that the combustion starting system does not draw air, oil, or mist out of the gas turbine engine air vents, or in any way upset the gas turbine engine bearing cavity air pressurization and venting.

g. Gas turbine engine fluid drains shall be located in relationship to the location of the combustion starting system such that they do not present a fire hazard.

h. If the combustion starting system exhaust is piped through the gas turbine engine enclosure, it shall be insulated and isolated such that it does not present a human safety hazard or a fire hazard. If the combustion starting system exhaust is interconnected to the gas turbine engine exhaust, the exhaust duct run shall be designed to prevent back flow of gas turbine exhaust into the combustion turbine when the combustion turbine is not operating. The exhaust design shall also prevent back flow of the combustion turbine exhaust into the gas turbine engine when the combustion turbine is operating. An automatic alarm system shall be installed to sense exhaust gas recirculation.

i. If the combustion starting system gas turbine draws its combustion air directly from the enclosure of the gas turbine engine it is dedicated to start, the combustion start system gas turbine shall have its own dedicated foreign object damage (FOD) protection screen.

j. A dedicated removal path shall be provided for the removal and replacement of the combustion start system gas turbine.

k. The combustion starting system gas turbine shall provide a minimum of 3,000 complete engine starts. Start life shall not be impacted by gas turbine engine motoring or water washing of the gas turbine engine it is dedicated to start, wash, and motor.

l. The combustion starting system shall not impact the effectiveness of the gas turbine engine enclosure firing fighting system.

3.4.9 Electrical system for gas turbine engines. Electrical equipment that connects to ship service power shall meet the power system interface requirements of MIL-STD-1399-300-1.

3.4.9.1 Electromagnetic interference (EMI) and electrical bonding and grounding. The gas turbine engine package shall be bonded and grounded in accordance with MIL-STD-461, MIL-STD-464, and MIL-STD-1310 to minimize adverse EMI effects and to protect personnel from electrical shock. Where EMI filters are required, line-to-line filters shall be used. AC line-to-ground capacitance shall be in accordance with MIL-STD-1399-300-1 and MIL-STD-461.

3.4.9.2 Equipment and cable grounding. Equipment shall be grounded and bonded in accordance with MIL-STD-1310.

3.4.9.3 Equipment grounding. Grounds for shielded apparatus shall be arranged to prevent the shield from being used as a return conductor. A single point connected to the ship's hull shall be the only ground connection for the apparatus within the shield.

3.4.9.4 Cable grounding. Individual shields shall be insulated from each other, and each shield in multi-lead systems shall be grounded to a single point as follows:

a. Short lead lengths. Single-point grounding shall be used for electromagnetic compatibility when lead lengths are short.

b. Long lead lengths. When the ratio of cable length to wavelength for the highest frequency to be used in the system is greater than 0.15, single-point grounding becomes ineffective. Multipoint grounding shall be used at intervals of 0.15 wavelength. When grounding the shield at intervals of 0.15 wavelength is impractical, ground shields shall be used at each end of the cable. Where feasible, flexible cables vice rigid shall be utilized.

3.4.9.5 Cable spacing and shielding. Power cables or other cables carrying signal information below 300 kilohertz (kHz) should be installed in accordance with the cable spacing and shielding requirements of S9407-AB-HBK-010.

3.4.9.6 Electrical safety. Design and construction of transducers and other components mounted on the engine shall be such that all exposed metal parts and the chassis shall be at ground (ship's hull) potential at all times. Exposed cables or sensors shall be protected to prevent inadvertent damage by personnel.

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3.4.10 Inlet plenum and exhaust plenum for gas turbine engines. The gas turbine engine inlet plenum to engine interface shall be designed to minimize pressure losses at the aerodynamic interface plane of the gas turbine engine.

3.4.10.1 Air inlet distortion. The inlet plenum shall meet the gas turbine engine's required radial and circumferential distortion intensity requirements as measured per ASME MFC-26 and SAE ARP1420.

3.4.10.2 Inlet and exhaust ducting connection. The inlet and the exhaust ducting connections between the gas turbine engine enclosure and the ship's inlet and exhaust ducting shall be flexible connections. The inlet and exhaust ducting connections shall be designed to be serviceable at the intermediate level (I-level) (see 6.6.5).

3.4.10.3 Inlet screen. Each gas turbine engine shall have its own inlet FOD screen. The inlet FOD screen shall be corrosion-resistant and 6.35- by 6.35-millimeter (mm) (¼- by ¼-inch) mesh. The FOD screen shall be strong enough for the operating conditions expected. It shall be located immediately in front of the engine inlet housing. This FOD screen, on its supplied mounting, shall be free of any damaging aerodynamic or engine-induced mechanical vibrations, shall not itself induce undesirable vibrations on the engine, and shall contain no obvious stress raisers such as flattened or calendared wire. Brazing of FOD screen wire shall be prohibited. The FOD screen shall be designed to minimize the disturbance of airflow into the engine inlet. The inlet FOD screen shall be designed to allow for removal and replacement.

3.4.11 General requirements for when the gas turbine engine is included within a base and enclosure.

3.4.11.1 Subbase. When a gas turbine engine exceeds 680 kilograms (1,500 pounds) in weight or 213 centimeters (84 inches) in length, the gas turbine engine shall be mounted on a rigid subbase.

a. The subbase shall maintain alignment of components without permanent deformation when subjected to the accelerations of testing in accordance with MIL-DTL-901.

b. The subbase shall be rigid enough to prevent misalignment of the attached units when the subbase is only supported at three extreme corners.

c. No additional bracing or stiffening of the subbase shall be required for handling or installation.

d. All components of the engine shall be mounted on the subbase except that the engine controller and the lubricating oil storage and conditioning assembly may be mounted separately from the subbase.

e. No part of the engine shall extend below the lower face of the subbase. The engine alignment support system and subbase assembly shall incorporate a means for realigning the engine to the driven equipment.

3.4.11.2 Alignment. Alignment of the engine and driven components of a unit shall be maintained by flanges, dowels, keys, thrust plates, or fitted bolts. Attachments for mounting of the engine shall permit the engine to move with thermal expansion. If used, metal expansion joints (flexible metal piping) shall be in accordance with MIL-DTL-17813.

3.4.11.3 Dimensions and weight. The items to be included in the dimension and weight determination shall include the engine, driven components, service fluids, attached piping, accessories, enclosure and subbase (if provided).

3.4.11.4 Inclination. The engine and accessory components, including piping systems, shall be designed to operate without loss of oil or cooling water under the following conditions:

a. Permanent trim up to 5 degrees on either end from the normal horizontal plane (temporary trim up to 31 degrees for amphibious ships).

b. Permanent list up to 15 degrees to either side of the vertical (31 degrees for amphibious ships).

c. Momentary trim of 10 degrees up or down from the normal horizontal plane for 10 seconds.

d. Momentary list up to 45 degrees to either side of the vertical for 10 seconds.

3.4.11.5 Piping and interfaces. Gas turbine engines shall be equipped with the accessories, equipment, or engine-attached piping necessary for gas turbine engine operation. This shall include but not be limited to valve fittings and flanges, an alarm system, and wiring or mechanical linkages required to connect detached accessories, instruments, and controls to the engine or to the cooling water, fuel, starting air, electrical, exhaust, or lubrication systems.

a. Gas turbine engine flexible fuel and oil system piping shall be in accordance with SAE AS150 or equivalent approved by the Technical Authority.

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b. Rigid piping shall be used where possible and shall be designed to minimize vibration or deflection without employing flexible hoses.

c. Other piping, both rigid and flexible, shall be in accordance with ASME B31.1 or equivalent approved by the Technical Authority.

d. Piping shall be securely strapped and supported by the engine structure and, if necessary, isolated by flexible hose to minimize the transmission of vibration, structure-borne noise, and shock loads. Flexible hose shall be mounted and clamped in a way that eliminates the possibility of abrasion by adjoining components or structures under vibratory conditions.

e. Tapered pipe thread connections shall not be used. All other connections shall be with flareless connections in accordance with MIL-DTL-18866 or welded and bolted flange connections.

f. Piping connections that interface between the gas turbine engine base (if applicable) and the ship, boat, or other craft shall have flanges in accordance with MIL-STD-777.

3.4.11.6 Flexible hose connections. Flexible hose shall be polytetrafluoroethylene (PTFE) in accordance with MIL-DTL-25579 or flexible metal hose in accordance with SAE AS1424. PTFE hose shall comply with SAE AS1072 for fire sleeves on all flammable fluid flexible hose. The fire sleeve shall extend over the hose socket and shall be securely clamped.

3.4.11.7 Cooling water. As specified (see 6.2), the gas turbine engine manufacturer shall define any cooling water requirements for power augmentation, emissions reduction, lubrication oil cooling, and water washing, subject to Technical Authority review.

3.4.12 Recuperators and intercooler heat exchangers. Recuperators shall meet the requirements of ASME Boiler and Pressure Vessel Code (BPVC) and shall be of the primary surface design. Compressor intercooling heat exchangers shall comply with API STD 616.

3.4.13 Control system. A dedicated digital control system shall be provided for each gas turbine engine. The control system shall provide for gas turbine engine starting and stopping, acceleration and deceleration, speed regulation, load acceptance and rejection, bleed air control (if so equipped), monitoring of system indicators, and prevention of surge and instability at all power levels and potential transient conditions. Each gas turbine engine control shall include all sensors and actuators necessary to perform the above control functions. Control shall include an emergency shutdown capability that can be operated automatically by failure detectors or that can be operated manually and independently of the control system. The control system shall meet the cybersecurity requirements of MIL-DTL-32613, and additional requirements as follows in 3.4.13.1 through 3.4.13.19.

3.4.13.1 Human machine interface (HMI). The control system shall include an HMI in accordance with MIL-DTL-32613 and MIL-STD-1472, consisting of a screen and input device to allow maintainers to view engine parameters in real time, update control software, create data trending, and perform troubleshooting. An inoperative HMI shall not inhibit control and operation of the gas turbine engine.

3.4.13.2 Reliability. The gas turbine control system shall have an MTBF in accordance with section 3.4.1.3. A control redundancy failure that permits continued operation of the gas turbine shall not count against the overall MTBF.

3.4.13.3 Control system functions. The control system shall include the following functions:

a. Sensor signal conditioning.
b. Control signal processing.
c. Output of control signals to actuators.
d. Warning and alarm processing.
e. Automatic limiting to prevent turbine damage.
f. Automatic tripping to prevent turbine damage.
g. Local control and indication.
h. Input and output interface.
i. Automatic start and stop sequencing.
j. Control gas turbine speed, power, or both in response to demand signals.
k. Load and speed regulation.

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l. Overspeed protection.
m. Compressor surge control.
n. Coordination between gas turbine control and generator governor control when driving a generator.
o. Hard wired stop button.
p. Dual redundant hard wired overspeed trips.
q. Communication to a remote monitoring and control station.

3.4.13.4 Accuracy. After the system has been calibrated, error introduced in signals by cabling and electronics shall be less than or equal to 1 percent of the signal range. Electronics recalibration intervals shall be 2 years or longer. Electronics error shall not exceed 2 percent of the signal range during the recalibration interval.

3.4.13.5 Software. Microprocessor and application programs shall conform to the standards, conventions, and procedures detailed in ISO/IEC/IEEE 12207 and ISO/IEC-IEEE 15288 or equivalent approved by the Technical Authority.

3.4.13.6 Reserves. The control system shall be designed and constructed for post-development growth in the requirements and use of the unit. In particular, 20 percent execution time reserves shall be provided in each execution from function of the program. Twenty percent memory reserves of each type shall be provided at delivery. Extra module slots, power, and cooling shall also be provided so that additional modules (cards) can be added in the ratio of one additional card to every four cards of each type originally installed, with a minimum of one of each type.

3.4.13.7 Communication. The control unit shall be designed to communicate sensor information and receive remote control commands through a digital serial data channel. Communications shall use redundant channels.

3.4.13.8 Built-in test (BIT). Fault detection and isolation shall be provided using BIT (or equivalent) and minimal operator intervention. The BIT (or equivalent) shall identify failures and faults in the following:

a. Sensors: sensor faults, indicating when signal values are going out of normal or defined limits.
b. Processors: power supply faults, memory faults, peripheral input/output faults, and faulty program sequences.
c. Communication: detection of physical communication breaks, excessively high error rates.

3.4.13.9 Power supply. Controllers shall be designed to operate with a primary and an emergency source of input power provided by the ship (see 6.2). The primary and emergency sources of power shall operate as follows:

a. Primary power shall be taken directly from the ship's power distribution system.
b. Emergency power shall be provided from an uninterruptible backup system, which shall:
(1) Be capable of starting the unit 1 hour after continuous operation without input power.
(2) Operate automatically independent of controller and not affect normal operations of the unit, including being capable of switching between primary and emergency sources with no impact to operations.
(3) Be capable of recharging to 50 percent capacity within 5 hours of input power being restored once the unit has been discharged for 1 hour.
(4) Include batteries that are separate from the charging system and replaceable.
(5) Protect the system's batteries from undercharging and overcharging.
(6) Have vents for the batteries of the system.
(7) Provide connection points for a non-permanently installed power supply to be tied in in a "dark ship" (complete loss of ship's power) scenario and provide control system electrical power for generator units such that recovery and restart is achievable during dark ship conditions.

3.4.13.10 Vibration monitoring. A vibration monitoring system shall be provided and mounted in an accessible location. A separate vibration signal conditioning unit shall be provided and mounted in an accessible location. The unit shall condition signals from the engine accelerometers, filter the signals at the rotating element speeds, and produce digital vibration amplitude signals to be transmitted to the control system. Buffered wide-band vibration signals shall be forwarded to the condition monitoring equipment. Buffered signals for wide-band acceleration, wide-band velocity, filtered vibration, and engine speeds shall be provided at the vibration signal conditioning unit for monitoring portable vibration processing or recording equipment. The vibration monitoring system shall provide

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both once per revolution and broad band vibration data outputs. The vibration standards of ISO 20816-3, modified if necessary (see 6.2) by the Technical Authority, shall be applied.

3.4.13.11 Surge monitoring. The gas turbine control system shall be designed to prevent compressor surge.

3.4.13.12 General instrumentation. The gas turbine engine shall have instrumentation to measure the parameters needed to perform control and diagnostics, to monitor the condition of the engine, and support the control functions identified herein. These parameters include, but are not limited to:

a. Rotating speed of each rotor system.
b. Output turbine/power turbine temperature. Temperature shall be both an average of all sensors and an individual readout of each measuring sensor.
c. Lubricating oil pressure.
d. Lubrication oil temperature.
e. Fuel manifold pressure.
f. Engine gas path pressures.
g. Engine gas path temperatures.
h. Vibration (accelerometers).
i. Fuel filter differential pressure.
j. Lubricating oil differential pressure.
k. Internal controller temperature monitoring.

3.4.13.12.1 Additional specific instrumentation. In addition to the above parameters, gas turbine engines shall have instrumentation to measure, as a minimum, the following additional parameters:

a. Variable stator vane position (where applicable to the gas turbine engine design).
b. Individual lubricating oil sump temperatures.
c. Compressor inlet temperature.
d. Compressor inlet pressure.
e. Power turbine inlet temperature.
f. Power turbine inlet pressure.
g. Compressor discharge pressure.
h. Inter compressor pressure and temperature (where applicable to the gas turbine engine design).
i. Compressor discharge temperature.
j. Engine exhaust temperature.
k. Power level position.
l. Fuel flow.
m. Variable turbine nozzle position (where applicable to the gas turbine engine design).
n. Enclosure temperature (when an enclosure is provided).
o. Lubricating oil scavenge and supply pressures and temperatures.
p. Start system parameters.
q. Controller temperature.
r. Accumulated gas turbine engine running hours.
s. Accumulated gas turbine engine starts. Start counters shall differentiate between normal starts and emergency starts (if provided).

3.4.13.13 Signal parameter warning. The control system shall provide a warning of unsafe conditions for, as a minimum, the following parameters:

a. Low lubricating oil pressure.
b. High lubricating oil temperature.
c. High turbine gas temperature.
d. Lubricating oil filter element high differential pressure.
e. Fuel filter element high differential pressure.

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f. High vibration (amplitude).

g. Sump high oil temperature alarm (7460 kW [10,000 shp] and larger only).

h. Low fuel-manifold pressure.

i. Low compressor discharge pressure, variable stator vane position (if applicable), and other parameters needed to preclude compressor stall.

j. Indication of compressor stall (7460 kW [10,000 shp] and larger only).

k. High coolant temperatures.

l. Low oil level on all type A applications.

m. High rotor speed.

n. Torque limit exceeded.

o. Others recommended by the engine manufacturer and approved by the Technical Authority.

3.4.13.14 Automatic limits and shutdowns. The control system shall control and limit the engine operation to prevent the degradation of hot section hardware and rotating parts from reducing the full service life of the engine. As a minimum, the gas turbine controller shall provide automatic shutdowns for overtemperature and overspeed conditions. All shutdown limits, with the exception of the overspeed shutdown, shall be battle over-rideable.

3.4.13.15 Overspeed protection. The engine shall have an overspeed protection. The controller system shall have a redundant overspeed feature. The redundant overspeed protection feature shall operate as follows:

a. The redundant overspeed protection device shall be capable stopping the engine at 110 percent of rated speed.

b. The redundant overspeed protection device shall operate independently of the primary control devices.

c. The primary control device shall not be able to influence the protection device's operation.

d. The redundant overspeed protection device shall be designed to operate using the power provided by the primary and emergency power supplies.

e. Normal gas turbine engine acceleration shall not activate the redundant overspeed protection device.

f. The redundant overspeed protection device shall work in series with all shutdown logic, including hardwired ship's control system, local stop pushbutton, and the primary controls.

3.4.13.16 Governing systems. When the gas turbine engine drives a generator, the governing system shall prevent surge and instability at all power levels and potential transient conditions due to changes in load. The engine shall be equipped with a primary and a backup governing system.

3.4.13.17 Emergency start capability. Emergency starting is defined as either a more rapid start than a normal start with the same or different start system than used for normal starting. When specified (see 6.2), the gas turbine engine start system control function shall provide capability for rapid starting of the gas turbine to support emergency start capability.

3.4.13.18 Fuel shutoff valves. Control of the valves shall be as follows:

a. The valves shall be capable of being closed by all shutdown logic as well as the local and remote stop pushbutton and the hardwired ship emergency stop pushbutton.

b. The valves shall be normally closed, power to open.

c. The valves shall shut within 1 second when commanded shut or when power is removed.

3.4.13.19 Controller location. When a gas turbine subbase is provided, the gas turbine engine controller shall be mounted so as to minimize heat absorption from operating machinery including the gas turbine itself.

3.4.14 Condition monitoring. When specified (see 6.2), the gas turbine engine shall be equipped with a condition monitoring and engine health assessment system. This system shall be in accordance with ISO 19860 and the general guidelines of ISO 17359 and SAE ARP1587. The condition monitoring system shall include display and input/output devices to enable the Organizational level (O-level) maintainer (see 6.6.10) to request and obtain engine diagnostic and prognostic information. The following capabilities shall be included in the condition monitoring system:

a. Continuous sensor validation. Sensor validation shall distinguish between an actual sensor failure and an actual engine alarm signal.

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b. Performance mapping of compressor surge, temperatures and pressures versus load, and specific fuel consumption versus load.

c. Identification of maintenance requirements for the turbine and its support equipment based on equipment condition as determined by software diagnostics.

d. Performance of data trending and diagnostics on current as well as historic data while continuing online monitoring.

e. Operation and maintenance logging to permit the operator to enter a maintenance activity or operating event manually.

f. Software flexibility that allows the system to continue to run while performing diagnostics with historic data.

g. Condition monitoring software shall allow for adding, deleting, and modifying diagnostics and diagnostic responses without changing the software. The capability shall be provided for designated personnel to create or modify diagnostics involving any combination of monitored parameters by using user-friendly interfaces, such as conditional tests that input into logic block diagrams.

3.4.14.1 Display. The condition monitoring system shall calculate, record, trend, and display the monitored data and the following:

a. Engine serial number.

b. Gas generator turbine starts and operating hours.

c. Power turbine starts and operating hours, if separable from gas generator starts and hours.

d. Enclosure operating hours.

e. Start time.

f. Compressor pressure ratio (compressor discharge pressure/compressor inlet pressure).

g. Estimated compressor and turbine efficiency.

h. Percent operating time between 0 and 25 percent power, between 26 and 50 percent power, between 51 and 75 percent power, and between 76 and 100 percent power.

3.4.15 Station numbering. Engine station numbering used in control, monitoring, and performance description shall comply with SAE AS755 and table III.

TABLE III. Engine station numbering.

Engine station numbering	Nomenclature
0	Conditions in free stream (inlet)
1	Duct/enclosure inlet flange interface
2	Compressor air inlet plane/face
2 1/	Low pressure compressor exit/high pressure compressor inlet (if applicable)
3	Compressor air exit plane/face
3 1/	Combustor inlet
4	Combustor exit/first turbine nozzle inlet face
4 1/	High pressure turbine exit/low pressure turbine inlet (if applicable)
5	Last gas generator turbine exit/free turbine inlet
5 1/	Free turbine/last turbine exit face
6	Not applicable
7	Gas turbine/diffuser nozzle interface plane
8	Exhaust diffuser/plenum exit plane

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TABLE III. Engine station numbering – Continued.

9	Duct/enclosure exhaust flange interface
10	Conditions in free stream (exhaust duct)

NOTE:
1/ An intermediate station within the gas turbine engine.

3.4.16 Smoke and emissions. The opacity of the exhaust smoke shall be no more than 20 percent when measured in accordance with 4.7.5. In addition, no white smoke shall be visible under any operating condition, including idle operation.

3.5 Maintenance and handling.

3.5.1 Shipboard maintainability. The gas turbine engine shall be designed to accommodate maintenance on, or removal of the following items, by Organizational-level maintainers without removal of the engine:

a. Fuel nozzles.
b. Igniters and excitors.
c. Starters, start valve, and general start system components.
d. Controls.
e. General pumps and valves.
f. Sensors.
g. Fuel manifold, fuel nozzles, fuel valves, fuel control components, and general fuel system components.
h. Lubrication pumps, filters, and general lubrication system components.
i. General piping.

3.5.2 Barring device. For engines with a free power turbine, a manual gas turbine engine barring device shall be provided to rotate the engine during layup and to prevent engine rotation during maintenance. For a single spool gas generator, the barring device shall not require removal of any components other than a protective cover over the barring tool installation point. For a multi-spool gas generator, removal of the inlet screen is acceptable.

3.5.3 Compressor cleaning. The gas turbine engine shall be provided with a means for compressor cleaning capable of removing oil, industrial fouling, and salt. Cleaning shall not require removal of any components, such as fuel or lubricating oil lines, accessories, or inlet bellmouths. Normal cleaning shall be accomplished during motoring mode operation. Cleaning with the engine online is acceptable as an enhanced wash capability, but is considered a backup to offline/motoring washing.

3.5.4 Borescoping. Engines shall incorporate a means to borescope, at a minimum, the compressors, combustors, and gas generator turbines. For multi-spool gas turbine engines, a means shall be provided for rotating each spool individually for borescoping.

3.5.5 Lifting means. Eyebolts, holes, or equivalent means shall be provided to permit lifting the complete engine and base enclosure, if provided, as a unit (including driven components and attached accessories) for general handling and installation without jacking the engine.

3.6 Configuration identification.

3.6.1 Model designation. Each gas turbine engine (including gears if supplied), generator set, or other complete gas-turbine-engine-driven package shall be assigned a unique model designation to completely identify the unit. Any changes or modifications, whether internal or external, that affect the form, fit, control interface, or functional interchangeability of the unit or require different logistics support shall require a suffix to the basic model designation or a new model designation to indicate the change.

3.6.2 Marking.

3.6.2.1 Marking of components and parts. Components and parts shall be marked for identification in accordance with MIL-STD-130. Repair parts shall be marked with the manufacturer's part number and serial

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number by casting, forging, stamping, or etching. When it is impractical to mark repair parts by any of these methods, the parts shall be identified by tags or a printed sheet on or in the package.

3.6.2.2 Engine identification. Engines shall be provided with an identification plate secured to the engine in a visible convenient location. Engine identification plates, as a minimum, shall include:

a. Engine type.
b. Serial number.
c. Model number.

3.6.2.3 Component identification. Each component, such as pumps, coolers, governors, instruments, starters, and sub-assemblies, shall have an identification plate. The identification plate shall be mounted where it can be easily read without danger to personnel. This plate shall conform to MIL-DTL-15024, type A, with the exception that aluminum alloys are not permitted. The plate shall contain the following information:

a. Manufacturer's name.
b. Model or part number.
c. Serial number.

3.6.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength. Parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with respect to installation and performance, except that matched parts or selective fits are permitted when identified as such. A new part number shall be assigned when a part or assembly is changed in a way that causes any of the following conditions:

a. Condition 1. Performance or durability is affected to such an extent that superseded items shall be discarded for reasons of safety or malfunction.
b. Condition 2. Parts, subassemblies, or complete articles are changed to such an extent that the superseded and superseding items are no longer directly and completely interchangeable with respect to installation and performance.
c. Condition 3. When repairable parent assemblies are interchangeable but contain non-interchangeable items, both the non-interchangeable items and the parent assemblies shall have unique identification numbers.

3.6.3.1 Part number. When a part or assembly is changed in such a way that none of the conditions specified in 3.6.3 occur, the part numbers shall remain unchanged. Under no condition shall the number be changed when a new use is found for an existing part. When an item has been furnished to the user, the applicable part number shall remain unchanged unless one or more of the conditions of 3.6.3 apply.

3.7 Gas turbine base and enclosure (also known as module). The gas turbine engine shall be mounted on a subbase structure (see 3.4.11) and housed within an enclosure, to be provided when specified (see 6.2), typically for all gas turbines on vessels of 500 displacement tons or more. The base and enclosure assembly (see 6.6.3) shall meet the requirements specified herein and as follows.

3.7.1 Equipment enclosure. The gas turbine enclosure assembly shall completely enclose the gas turbine engine and its accessories.

3.7.2 Strength and durability. The gas turbine base and enclosure shall be designed so that:

a. The base strength and rigidity shall maintain the gas turbine engine alignment between the gas turbine engine to base mounting points, thereby preventing misalignment of the engine between the mount points.
b. The roof of the gas turbine enclosure shall be strong enough to allow up to 3 persons to work on top of the enclosure.

3.7.3 Materials. All materials shall be resistant to or protected to resist corrosion and comply with 3.2. All valves mounted inside the gas turbine enclosure shall be made of corrosion-resistant steel.

3.7.4 Access doors and windows. The gas turbine enclosure shall have access doors and viewing windows for easily inspecting, maintaining, and replacing the engine and other interior equipment. An access door/hatch shall be provided to permit access to either side of the gas turbine within the base/enclosure. The access doors may contain viewing windows. The viewing windows shall be placed so the operators can easily inspect the oil lines for leakage. An access door shall be located adjacent to the gas turbine compressor, and a viewing window shall allow a clear

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sight of the gas generator. All viewing windows shall be shatterproof and scratch-resistant. The gas turbine enclosure, where feasible as allowed by the ship installation layout, shall have viewing windows on at least two sides. Large (i.e., step-in) enclosures shall have a secured, but removable, ladder within the module that leads to the roof access if a roof access is provided with the enclosure.

3.7.5 Safety. Fuel oil, lubricating oil, and other combustible fluid lines and fittings shall be isolated to prevent leaks on any component that could pose a fire hazard.

3.7.5.1 Surface temperature. The interior of the gas turbine enclosure walls shall be insulated so that its exterior surface temperature is no higher than 52 °C (125 °F). The thermal insulation shall be protected from oil, grease, moisture, dirt, and other contaminants and from physical damage. The thermal insulation may also be part of the acoustic barrier required by section 3.7.11. For gas turbine engines that exceed 7,460 kW (10,000 shp), interior surfaces that could have a temperature of 204 °C (400 °F) or higher shall be insulated or shielded if a flammable fluid could impinge on the surface, with the exception of the gas turbine engine combustor and turbine cases.

3.7.5.2 Warning information. A sensor shall be provided to output the internal module temperature to the engine control system. A warning sign shall be placed near each module access door that specifies the cooldown time (time required for the interior walls of the module to reach 52 °C [125 °F] or lower) as a function of power level. This warning sign shall include safety precautions to be followed if immediate entrance to the module is required after gas turbine engine shutdown.

3.7.5.3 Flanges. Flanges shall be equipped with flange shields to prevent the leakage of flammable fluids from contacting hot surfaces.

3.7.5.4 Guards. Guards shall be provided to protect personnel from hazards such as exposed rotating shafts or pulleys, high voltages, belts, chains, and any other hazardous condition.

3.7.5.5 Fire sleeves. Fire protection sleeves shall be in accordance with 3.4.11.6 and shall be used on all non-metallic flexible hoses.

3.7.6 Subbase. Unless otherwise specified (see 6.2), where the driven equipment is a generator, the gas turbine engine and the generator are to be mounted onto the same subbase.

a. When the gas turbine engine manufacturer is not the source of the subbase, the gas turbine engine manufacturer shall verify subbase compliance with the gas turbine manufacturer’s requirements.

b. The subbase shall be of sufficient strength to prevent misalignment of the attached units when the subbase is rigidly supported at three extreme corners.

c. All components of the gas turbine engine shall be mounted on the subbase. The gas turbine engine controller, the gas turbine lubricating oil and conditioning assembly, and the hydraulic starting unit may be mounted external to the subbase.

d. The gas turbine enclosure walls shall be rigidly attached to the subbase.

e. No part of the gas turbine engine assembly, including heat exchangers, shall extend below the lower face of the subbase.

3.7.7 Mounts.

3.7.7.1 Resilient mounts. The supports shall be designed so that the load is equally distributed to each mount. Resilient mounts shall be in accordance with S9073-A2-HBK-010.

3.7.7.2 Shock mounts. The resilient mounts and the shock mounts may be the same mount or mounts of a separate design.

3.7.8 Air inlet system. The gas turbine enclosure air inlet or inlet plenum shall be connected to the gas turbine engine air inlet (bellmouth) through a removable, watertight, and airtight flexible connection. The gas turbine enclosure air inlet or inlet plenum shall be separated from the part of the gas turbine enclosure module that encloses the gas turbine engine by a watertight and airtight barrier. The flexible connections and barrier shall be designed so they can be easily removed and replaced.

3.7.8.1 Airflow distortion. The gas turbine enclosure air inlet or inlet plenum shall be designed to minimize airflow distortion to the gas turbine engine. The gas turbine enclosure inlet or inlet plenum, when coupled with the

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ship gas turbine air intake, shall be designed to meet the airflow quality requirements specified by the engine manufacturer. Flow control devices, such as swirl vanes and inlet scrolls may be used, provided they do not interfere with the routine inspection of the inlet or inlet plenum, the inlet screen, or the gas turbine bellmouth area and do not interfere with engine removal.

3.7.8.2 Inlet screen. A screen to protect the gas turbine engine from foreign object damage shall be installed directly in front of the gas turbine engine inlet bellmouth in accordance with 3.4.10.3.

3.7.8.3 Rivets and potential foreign objects. Rivets and repair or patching compounds shall not be used in the inlet or inlet plenum.

3.7.8.4 Inlet system connection. Flexible joints shall be used to connect the gas turbine enclosure inlet with the ship intake duct. This connection shall be designed so it can be easily replaced.

3.7.9 Exhaust system. The gas turbine enclosure exhaust or exhaust plenum shall be designed to minimize pressure loss while maintaining enough clearance to allow personnel access for visual inspection of the output turbine last-stage blading. Flow control devices, such as turning vanes or rolls, may be used. The materials used in the module exhaust or exhaust plenum shall be structurally strong enough to withstand maximum operating temperatures under normal operating pressure and designed to withstand the corrosive environment of a gas turbine exhaust.

3.7.9.1 Exhaust system connection. Flexible joints shall be used to connect the gas turbine enclosure exhaust with the ship exhaust duct. This connection shall be designed so it can be easily replaced.

3.7.10 Cooling system. The gas turbine enclosure shall interface with the ship cooling air supply system. The cooling air supply system shall protect the engine and engine-mounted accessories and components from high temperatures during operation and after engine shutdown on the basis of a maximum inlet air temperature of 38 °C (100 °F). The cooling airflow shall enter the module through ventilation openings and be directed around the engine toward the exhaust duct.

3.7.10.1 Ventilation damper. The cooling air entry shall have a ventilation damper. The cooling air damper tightness requirements shall support the requirements of the fire protection system. The damper shall be automatically actuated by the gas turbine engine control system. The damper shall also have manual controls for use by operators. The damper shall also be automatically activated by the fire protection system.

3.7.10.2 Cooling system connection. Flexible joints shall be used to connect the gas turbine enclosure cooling air supply with the ship cooling air duct. This connection shall be designed so it can be easily replaced.

3.7.11 Acoustic requirements. The module wall and roof panels shall be designed and constructed so that the enclosed engine meets the airborne noise requirements (see 3.7.23). Sound-absorbing materials shall be protected from absorbing oil, grease, dirt, moisture, or other contaminants and from physical damage. Sound-absorbing materials shall be fire-resistant and self-extinguishing and shall not release toxic gases during combustion. The acoustical materials may also serve as part of the thermal barrier (see 3.7.5.1).

3.7.12 Power takeoff seal. A flexible cover between the gas turbine enclosure output shaft to the driven load shall be provided.

3.7.13 Enclosure drainage. The gas turbine enclosure drainage systems shall be designed to drain the enclosure regardless of the enclosure rake angle within the ship. The gas turbine enclosure and inlet drainage shall be separated to prevent the gas turbine inlet suction drawing drainage from the enclosure drainage system.

3.7.14 Enclosure lighting. The gas turbine enclosure shall contain a lighting system that provides illumination of no less than 301 candela (28 footcandles) throughout the enclosure interior. The lighting system shall be explosion-proof. Shatterproof light assemblies capable of withstanding the gas turbine enclosure environmental conditions for a minimum of 5 years under normal operating conditions shall be provided.

3.7.15 Anti-icing detector. An anti-icing sensor shall be provided inside the gas turbine enclosure air inlet or inlet plenum. The sensor shall provide an input to the ships gas turbine intake anti-icing system to permit actuation of ship's anti-icing system when the temperature of the intake air is 4 °C (40 °F) or lower, and the relative humidity of the intake air is above 70 percent.

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3.7.16 Piping systems. Piping and connectors shall be designed and constructed in accordance with 3.4.11.5. In addition, piping located on the module floor, which may be damaged by personnel while working in the module, shall have guards installed for protection.

3.7.17 Gas turbine enclosure connections. All connections (for example, electrical, pneumatic, fuel lubrication, firefighting, and water) into the gas turbine enclosure shall be made through the subbase. Piping connections that interface with ship piping shall be in accordance with MIL-STD-777.

3.7.18 Fuel systems. Metallic fuel piping shall be corrosion-resistant steel. Flexible hoses shall be in accordance with MIL-DTL-25579. Snap rings shall not be used in the fuel system.

3.7.18.1 Enclosure heater. For a gas turbine engine 7,460 kW (10,000 shp) or larger, a heater inside the module shall be used to maintain the fuel at a temperature that prevents any fuel constituents from precipitating. The heater shall be used during the engine prestart and start period in addition to “cold iron” maintenance operations.

3.7.19 Lubrication systems. Material containing more than 5 percent copper shall not be used in the off-engine-mounted lubricating oil system. Snap rings shall not be used in the lubrication system. Flexible hoses shall be in accordance with MIL-DTL-25579. Shutoff valves, other than manually operated valves, shall fail-safe close when de-energized.

3.7.20 Starting systems. The gas turbine enclosure assembly shall be compatible with the engine starting system.

3.7.21 Electrical systems. Electrical wiring and power supplies that are located within the gas turbine enclosure shall be protected from heat and damage. Electrical and electronic equipment shall be in accordance with 3.4.9. Bonding and grounding of the gas turbine enclosure assembly and the electrical and electronic systems shall be in accordance with 3.4.9. Protective guards with a warning label plate shall be provided over 440 volt or greater terminals.

3.7.22 Fire protection systems. The gas turbine engine base enclosure assembly shall be equipped with a fire safety system consisting of a fire detection subsystem, visual and audible alarms, fire-extinguishing system interfaces and discharge piping, and control functions for fuel and cooling air. Firefighting agents shall be as specified (see 6.2), and may be Halon, HFC-227ea in accordance with ASTM D6064, water mist, or other firefighting mediums.

3.7.22.1 Fire detectors. The gas turbine engine base enclosure shall be equipped with infrared or a combination of ultraviolet and infrared flame detectors and fixed temperature fire detectors. Each detector type shall provide full coverage of the interior module space. The detectors shall be in accordance with NFPA 72. The detectors shall be able to operate continuously at the maximum internal temperature in the gas turbine engine base enclosure without any damage, malfunction, false alarm, or trouble indication. The detectors shall withstand the requirements of this specification without any damage, malfunction, false alarm, or trouble indication.

a. The flame detectors shall be able to detect a 0.1 square meter (1 square foot) JP-5 flame at a distance not to exceed 12 meters (40 feet) within 1 second of full ignition.

b. The flame detectors shall be cross-zoned so that every point of the module assembly shall be monitored by two detectors on separate zone circuits with different viewing angles. Operation of the first detector shall cause a pre-alarm or trouble signal. Operation of a detector on a second zone circuit shall close the contacts at the fire detection control panel and initiate the firefighting agent discharge, close the ventilation dampers, close the fuel lines, and perform other functions as defined herein.

3.7.22.2 Alarm devices. Audible and visual fire alarm devices shall be provided on the interior and exterior of the module. The devices shall be used to warn personnel of fire detection and firefighting agent discharge.

3.7.22.3 Inhibit switches. Inhibit switches shall not be provided.

3.7.22.4 Fire detection control unit. When specified (see 6.2), a fire detection control unit shall be provided as part of the gas turbine unit. The fire detection control unit shall comply with UL 864 and shall continuously monitor the detection and alarm circuits for open circuits, short circuits, grounds, contaminated lenses, or other conditions that would prevent normal operation of the system.

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3.7.22.4.1 Pre-alarm signal. The fire detection control unit shall provide a pre-alarm signal whenever a single flame detector zone circuit goes into the alarm condition. The pre-alarm signal shall:

a. Close a pair of normally open isolated contacts (used to provide a signal to the remote monitoring and control system).

b. Energize audible and visual signals at the control unit.

3.7.22.4.2 Fire alarm signal. The fire detection control unit shall be able to be configured to provide a fire alarm signal and an emergency stop command upon operation of any combination of heat detectors and flame detector zones. Upon activation, the control unit configuration shall provide the fire alarm signal whenever one of the following occurs:

a. Energize audible and visual signals at the control unit.

b. Energize the audible and visual fire alarm devices.

c. Energize a signal capable of driving an electric solenoid actuation device.

3.7.22.4.3 Firefighting agent actuated signal. The fire detection control unit shall provide a firefighting agent actuated signal whenever it receives a signal from a pressure switch located in the firefighting agent actuation system. The firefighting agent actuated signal shall:

a. Energize the audible and visual signals at the control unit.

b. Close the gas turbine base enclosure air ventilation dampers.

c. Shut down the cooling air fan.

d. Provide the gas turbine base enclosure firefighting agent release signal to the control system.

e. Energize the audible and visual fire alarm devices. The audible and visual alarms shall cycle on and off continuously until acknowledged.

3.7.22.4.4 Trouble signal. The fire detection control unit shall provide a trouble signal whenever the normal functioning of the fire detection system is interrupted, including:

a. Occurrence of a short circuit or open circuit on any enclosure fire detector circuit.

b. Contamination of any fire detector lens.

c. Closure of a pair of normally open isolated contacts (which shall be used to provide a signal to the remote monitoring and control system).

d. Energizing of an audible and visual signal at the control unit.

3.7.22.4.5 Visual indicators. The fire detection unit shall provide a visual indication of the base enclosure fire detector zone circuits that have gone into the alarm condition. Each zone's visual indication, once energized, shall remain energized until it is manually reset. Each fire detector that is not on an individual zone circuit shall have a visual indicator that is energized when the detector goes into alarm condition and remains energized until it is manually reset from the control unit.

3.7.22.5 Firefighting piping, valves, and fittings. The base enclosure's firefighting piping system shall include the discharge piping and nozzles within the base enclosure and the interface flange at the exterior of the base enclosure in the subbase. The piping and fittings shall be installed in accordance with S9074-AR-GIB-010/278. The piping, fittings, and flexible connections (if used) shall be in accordance with MIL-STD-777. Flexible connections (if used) shall also be in accordance with MIL-E-24572, if possible. Discharge piping shall be supported commensurate with high-pressure discharge piping.

3.7.22.6 Enclosure airtightness. The airtightness of the enclosure shall be designed to maintain the required firefighting agent concentration when all enclosure penetrations are sealed, drains are capped, and the module air inlet and cooling air openings are closed as follows.

a. Automatically operating dampers shall seal off the air inlet and cooling air inlet and shut off the cooling fan (when so equipped) when the fire protection system is activated in order to maintain the firefighting agent concentration inside the base enclosure.

b. The exhaust opening around the gas turbine exhaust plenum or eductor may or may not be closed off during the firefighting agent discharge and shall be accounted for in the concentration and discard calculations. If

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the cooling air is exhausted by the eductor effect of the engine exhaust through an annular opening, special design consideration may be required to meet the concentration requirements.

3.7.23 Engine accessibility and removal.

3.7.23.1 Equipment accessibility. The gas turbine engine base enclosure shall be designed with the required space around the gas turbine engine and equipment for performing routine inspection and maintenance without the removal of hardware. Where the gas turbine engine design accommodates in-place balancing, the enclosure shall have adequate space to accommodate the in-place balancing capability.

3.7.23.2 Equipment removal. The gas turbine enclosure shall have removable panels, large access doors, removal rails, or other design features for easily removing and replacing the engine and equipment mounted inside the enclosure.

a. For gas turbine engines larger than 12 megawatt (MW) (16,100 shp) power output, engine removal shall be through the gas turbine enclosure and ship's inlet duct.

b. When specified (see 6.2), rails, special tooling, or other apparatus shall be provided to guide the engine during removal and prevent damage to the engine, module, ductwork, or other shipboard equipment.

3.7.24 Noise.

3.7.24.1 Airborne noise. The airborne noise limit shall be in accordance with MIL-STD-1474 grade D, or as specified (see 6.2).

3.7.24.2 Structureborne noise. The structureborne noise limit shall be in accordance with MIL-STD-740-2, type III, or as specified (see 6.2).

3.8 Shock. Unless otherwise specified (see 6.2), the gas turbine engine and all systems and components of the module assembly shall be physically tested to grade A shock requirements of MIL-DTL-901, except that the gas turbine water wash system and gas turbine module heater shall meet grade B shock requirements.

3.9 Gas turbine engine generator set test (previously accepted gas turbine). When a previously Navy-accepted gas turbine engine with a mechanical propulsion rating is proposed for a generator set application, the previously approved gas turbine mechanical propulsion rating needs to be reduced by 10 percent to accommodate the ten percent electrical overload requirement of MIL-G-22077.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one production representative complete engine or engine package in accordance with table IV.

TABLE IV. First article and conformance verifications.

Requirement nomenclature	Requirement	Verification	First article		Conformance	
			Engine	Package (when included)	Engine	Package (when included)
DAP	3.1	4.4 and appendix A	X	X		
Rated power	3.1.1	4.6 through 4.6.6	X			
Fuel consumption	3.1.2	4.6 through 4.6.6	X			
Operability	3.1.3	4.7.9 and 4.7.10			X	
Materials	3.2 through 3.2.4 and 3.2.6	4.5	X	X	X	X

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TABLE IV. First article and conformance verifications – Continued.

Requirement nomenclature	Requirement	Verification	First article		Conformance	
			Engine	Package (when included)	Engine	Package (when included)
Health hazard materials	3.2.5	4.6.10	X	X		
Salt air environment	3.3.2	4.6.2.2	X	X		
Design life, HSRI, MTBF, and TARP	3.4.1.1 through 3.4.1.3 and 3.4.2	4.6.11 and appendix A, if DAP conducted	X			
Construction	3.4.3 through 3.4.4.2	4.5 and 4.7.10	X	X	X	X
Fuel system	3.4.5 through 3.4.5.10	4.5 and 4.6.4.1.6	X		X	
Lubricating oil	3.4.6 through 3.4.6.11	4.6.4.1.5 through 4.6.4.1.5.2	X	X	X	X
Bleed air	3.4.7 through 3.4.7.3	4.6.4.1.2	X	X	X	X
Starting	3.4.8 through 3.4.8.5	4.7.2.1	X		X	
Starting	3.4.8	4.7.2	X			
Emergency starting	3.4.13.17	4.7.2.3	X		X	
Electrical	3.4.9 and 3.4.9.2 through 3.4.9.6	4.5	X	X		
EMI	3.4.9.1	4.6.12	X	X		
Inlet and exhaust plenum	3.4.10.1 through 3.4.10.3	4.6.13	X	X		
Base and enclosure	3.4.11 through 3.4.11.3 and 3.4.11.6; 3.7 through 3.7.21	4.5 and 4.6.6.l	X	X		
Inclined operation	3.4.11.4	4.6.8 through 4.6.8.3, and 4.7.10	X	X		
Piping integrity	3.4.11.5	4.5 and 4.7.1	X	X	X	X
Cooling water	3.4.11.7	4.6.4.1.7	X			
Recuperators and intercoolers	3.4.12	4.5	X	X		
Control system	3.4.13 through 3.4.13.9, 3.4.13.11, 3.4.13.18, 3.4.13.19, and 3.4.15	4.6.1.3, 4.6.4.1.3, 4.6.4.1.4, 4.6.6; (or 4.7.9 for conformance)	X	X	X (4.7.9)	X (4.7.9)

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TABLE IV. First article and conformance verifications – Continued.

Requirement nomenclature	Requirement	Verification	First article		Conformance	
			Engine	Package (when included)	Engine	Package (when included)
Vibration monitoring	3.4.13.10	4.6.4.1.1	X		X	
Control instrumentation and shutdowns	3.4.13 through 3.4.13.14	4.6.4.3 (or 4.7.9 for conformance)	X (4.6 all)		X (4.7.9)	
Overspeed protection	3.4.13.15	4.7.4 and 4.7.10	X		X	
Governor operation	3.4.13.16	4.6.4.1.3, 4.6.4.1.4 (or 4.7.9 for conformance)	X (4.6 all)		X (4.7.9)	
Condition monitoring	3.4.14 and 3.4.14.1	4.6.4.1 and 4.6.4.1.3	X		X	
Emissions	3.4.16	4.6.4.1.8 and 4.6.9	X			
Smoke	3.4.16	4.7.5	X		X	
Maintenance and handling	3.5 through 3.5.4 and 3.7.23 through 3.7.23.2	4.6.11	X		X	
Configuration, designation, marking	3.6 through 3.6.3.1	4.5	X	X	X	X
Surface temperature	3.7.5.1	4.7.6		X		X
Fire protection system	3.7.22 through 3.7.22.5	4.7.8	X	X	X	X
Enclosure airtightness	3.7.22.6	4.7.7		X		X
Airborne noise	3.7.24.1	4.6.14	X	X		
Structureborne noise	3.7.24.2	4.6.15	X	X		
Vibration	3.4.3.14	4.6.17	X	X		
Shock	3.8	4.6.7 through 4.6.7.4	X	X		
New application generator set	3.9	4.6.16	X	X		

NOTE:

1. Package includes the engine, the enclosure (when included), the controls, and other associated components assembled and delivered for installation during new construction or major overhaul.

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4.3 Conformance inspection. Conformance inspection shall be performed on a complete engine or engine package in accordance with table IV.

4.4 Design assessment process examination. Where a DAP has been conducted (see 3.1, 6.2, and appendix A), the Technical Authority will make a determination of whether the engine is considered a new design, that an existing design may have endurance and other testing requirements adjusted, that an existing gas turbine of a particular navy type may be accepted for use as a different type, or that uprating is applicable.

4.5 Visual, dimensional, and materials examination. Each gas turbine or gas turbine package offered for delivery shall be examined for general conformance to requirements herein regarding materials, construction, fuel, lubricating oil, bleed air, starting, electrical, inlet and exhaust, control, and monitoring systems; marking, enclosure (if applicable) construction and fire protection, and any other applicable requirements specified herein not involving disassembly or tests. Noncompliance with any specified requirements shall constitute cause for rejection.

4.6 First article testing. First article gas turbines shall be subject to comprehensive testing to include operating endurance testing, to include first article pre-and post-test examinations and tests, first article specific tests (see table IV), shock testing (see 4.6.7), and general tests and examinations, some of which are also applicable to conformance testing (see 4.3).

4.6.1 First article pre-test examinations and tests. In addition to the examination of 4.5, gas turbine engines for which first article testing is required shall have the following conditions, tests, and inspections, conducted before commencing first article testing.

4.6.1.1 Component acceptance criteria and wear limits. The acceptance limit criteria for all wearing engine components, variability among the parts of the same part number, dimensional acceptance limits, variability limits, limits for all running clearances and adjustments, design tolerances and clearances, engine operating parameters, maximum loads for speeds between idle speed and rated speed, and lubricating oil spectrographic analysis limits shall be determined prior to the start of testing. At a minimum, the engine components noted in table V shall have design tolerances, clearances, and adjustments, and condemning wear limits shall be identified.

TABLE V. Engine component wear measurements list.

No.	Engine component	Measurement
1.	Axial compressor Rotor Casing Blade tip Blade	 Outer diameter (OD) Inner diameter (ID) Clearance Weight
2.	Centrifugal compressor	Clearance
3.	Combustor	Liner thickness
4.	High pressure turbine Rotor diameter Turbine tip	 OD Clearance
5.	High pressure turbine nozzle	Area
6.	Power producer turbine Rotor diameter Casing diameter Turbine tip	 OD ID Clearance
7.	Power producer turbine nozzle	Area
8.	Gear train	Backlash
9.	Gears Bushing	End-play ID
10.	Injector nozzle	Opening pressure, spray pattern, flow, leak down rate

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TABLE V. Engine component wear measurements list – Continued.

11.	Fuel pump	Flow
12.	Fuel flow throttling valve	Flow at specified command signal
13.	Lube pump	Flow
14.	Bearings Bearing housing	ID, OD Shaft to outer bearing clearance
15.	Seals	Clearance

4.6.1.2 Engine history. The previous operational history of the first article test engine shall be identified.

4.6.1.3 First article test control system. The gas turbine engine first article test control system differences from production engine control approach shall be identified.

4.6.1.4 Maintenance schedule. Gas turbine engine maintenance scheduled to be conducted during testing shall be identified prior to first article testing. Any redesign or modification of the proposed engine manufacturer's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet the specified requirements shall be identified (see 6.2).

4.6.1.5 Lubricating oil consumption and analysis. Anticipated lubricating oil consumption rates shall be determined for approximately each 250 operating hours, and baseline oil analysis determined, prior to commencement of first article testing.

4.6.1.6 Test site examination. Prior to commencement of first article testing, the site for the testing shall be examined for full capability to conduct the testing, to include monitoring and safety.

4.6.1.7 Performance parameters. Allowable vibration levels, specific fuel consumption, rated power, and turbine control temperatures expected shall be determined prior to commencement of first article testing.

4.6.1.8 Starter test. Prior to the commencement of endurance testing, the engine shall be subjected to 10 consecutive normal starts by the proposed start method and apparatus intended for end use, except for an auxiliary combustion engine-based start system, which shall be subjected to three consecutive normal starts by the proposed starting system. The time to start, time to reach minimum idle speed, minimum starting rpm, and minimum starter requirements shall be determined during these start tests. The time to start shall be the time required from initial engagement of the starter mechanism (the engine shall be at a standstill) to the time at which idle speed is attained. The starting source energy requirements shall be validated during these start tests (not required for an auxiliary combustion engine-based start system).

4.6.2 First article endurance test environmental conditions. The conditions of 4.6.2.1, 4.6.2.2, and 4.6.2.3 shall be in place during the conduct of the first article endurance testing.

4.6.2.1 Intake air depression and exhaust back pressure restriction settings. The intake air depression and exhaust back pressure restriction values specified in table I shall be set for all first article tests. The specified restriction values shall be set at corrected rated load and speed. The settings shall remain unchanged during all first article tests.

4.6.2.2 Salt injection requirements. The engine shall be tested with salt injection during the first article endurance test. A saltwater solution shall be introduced into the inlet air stream such that the concentration levels at the compressor inlet measurement plane are in accordance with table VI during the first article test. The salt solution shall be as follows:

a. $MgCl_2$ $6H_2O$: 11.0 grams.

b. $CaCl_2$ (anhydrous): 1.2 grams.

c. Na_2SO_4: 4.0 grams.

d. NaCl: 25.0 grams.

e. H_2O: To make 1 liter of solution.

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TABLE VI. Salt concentration at compressor measurement inlet plane.

Operating time, % of total	50	45	3	2
Total salt concentration (ppm by weight)	0.003	0.007	0.016	0.03
Droplet size range % of total ppm				
Above 13 micrometers (microns, µm)	25	45	45	50
4 to 13 µm	25	30	30	25
Less than 4 µm	50	25	25	25

4.6.2.3 Ambient operating conditions. The engine shall be subjected to the NSD test operating conditions in table I unless otherwise approved by the Technical Authority prior to the start of testing. Throughout the first article testing (especially at the 100 percent power point), the engine shall be operated at a mass average temperature at the inlet to the high-pressure turbine stator that would be experienced when operating at the NSD conditions specified in table I. If there is a deviation, then the test results shall be corrected to NSD condition based on the power rating correction factors given in ISO 2314 or other thermodynamic cycle model approved by the Technical Authority.

4.6.3 Fuel oil and lubricating oil requirements.

4.6.3.1 Fuel. The test fuel shall be naval distillate fuel in accordance with MIL-DTL-16884 (NATO F-76). Request to use commercially available alternative fuel during testing shall be subject to approval by the Technical Authority (see 6.2). Alternative commercial fuel shall meet the fuel property requirements listed in table VII.

TABLE VII. Fuel property requirements for alternatives to naval distillate fuel.

Characteristic	Requirement	ASTM test method	ISO test methods
Water & sediment, % volume (max.)	0.05	ASTM D2709 ASTM D6304	ISO 3733
Ash content, % weight (max.)	0.005	ASTM D482	ISO 6245
Carbon residue, on 10% bottoms, % weight (max.)	0.20 0.14 0.14	ASTM D524 [1/] ASTM D4530 ASTM D189	ISO 10370
Lower heating value, kJ/kg [2/]	42,800	ASTM D240	
Ignition quality, Cetane no. (min.) Cetane index (min.)	 42 43	ASTM D976 [3/]	 ISO 5165 ISO 4264
Particulate contamination, mg/l (max.)	10	ASTM D6217	
Sulfur content, % weight (max.)	0.5 [4/]	ASTM D4294 [5/]	ISO 8754 or ISO 14596
Viscosity at 40 °C, square millimeters/second	1.7 – 4.3	ASTM D445	ISO 3104
Lubricity, HFRR [6/] at 60 °C, micrometer (max.)	520	ASTM D6079	ISO 12156-1

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TABLE VII. Fuel property requirements for alternatives to naval distillate fuel – Continued.

NOTES:

1/ When the finished fuel contains a cetane improver, the carbon residue requirement specified shall apply to the base fuel without the cetane improver.

2/ Kilojoule per kilogram

3/ Either cetane number or cetane index (ASTM D976) shall be reported. The cetane index requirement specified shall apply to the base fuel without cetane improving additives. Where cetane index is reported, the value shall be reported as the cetane index.

4/ Any sulfur content up to the maximum is allowable.

5/ ASTM D1552 and ASTM D2622 are also acceptable methods of measuring sulfur content.

6 High frequency reciprocating rig (HFRR) lubricity is measured in micrometers, µm.

4.6.3.2 Lubricating oil. Lubricating oil used during the tests shall be the engine manufacturer's selected version of MIL-PRF-23699 or MIL-PRF-17331 lubricating oil in accordance with 3.4.6.2.

4.6.4 Conduct of endurance operational test. Unless otherwise specified (see 6.2), all type A engines shall complete 1,000 hours of endurance testing by repeating cycles accordance with table VIII. All type B engines shall complete 3,000 hours of endurance testing by repeating cycles in accordance with table IX.

a. These endurance testing hours may be reduced if so determined by the Technical Authority as part of the appendix A gas turbine engine design assessment process.

b. The test cycles of tables VIII and IX are also used for conformance testing with only one cycle required (see 4.7.9).

c. For each engine operating condition, with the exception of the idle operating condition, the corrected load shall be maintained within plus or minus 1 percent of the target load value, and the speed shall be maintained within plus or minus 1 percent or plus or minus 10 rpm (whichever is greater) of the target speed value.

TABLE VIII. Endurance test cycle for type A gas turbine engines.

Step	Load (percent)	Time (minutes)
1	Idle	2
2	100	10
3	20	5
4	Idle	3
5	60	10
6	100	20
7	90	10
8	0 (not operating)	60 1/

FOOTNOTE:

1/ Upon completion of step 7, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20%] may have to be reduced).

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TABLE IX. Endurance test cycle for type B gas turbine engines.

Step	Load (percent)	Time (minutes)
1	Idle	5
2	30	20
3	50	15
4	100	60
5	90	20
6	80	20
7	100	30
8	60	20
9	Idle	20
10	100	30
11	90	40
12	100	30
13	20	60
14	50	30
15	Idle	1
16	100	11
17	Idle	1
18	90	1
19	Idle	1
20	100	1
21	Idle	1
22	50	1
23	90	1
24	Idle	1
25	50	10
26	20	5
27	30	5
28	0 (not operating) 1/	60

FOOTNOTE:

1/ Upon completion of step 27, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20%] may have to be reduced).

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4.6.4.1 Monitoring and testing during endurance testing. Monitoring and testing conducted during endurance testing shall include vibration, bleed air, performance, instrumentation, lubricating oil, fuel oil, and cooling water. Other inspections, such as exhaust emissions, and testing required for first articles may be conducted during endurance testing if permitted by the Technical Authority.

4.6.4.1.1 Gas turbine engine vibration monitoring. While operating at maximum continuous speed and at other speeds specified in the test agenda in the operating speed range, sweeps shall be made for vibration amplitudes at frequencies other than synchronous. As a minimum, these sweeps shall cover a range of 0.25 to 8 times vane passing frequency. If the amplitude of any discrete, non-synchronous vibration exceeds 20 percent of the allowable vibration, the first article testing shall be halted until resolution of the source of the increase in vibration level. Plots showing vibration amplitudes and phase angle versus speed for acceleration and deceleration shall be made. The speed range covered by these plots shall be zero to the specified trip speed.

a. Vibration sensors. If the production unit does not have vibration sensors include in its design, vibration sensors shall be installed as part of first article testing.

b. Broadband vibration. Unfiltered vibration measurements shall be recorded throughout the operating speed range.

4.6.4.1.1.1 Critical speed verification. For variable speed gas turbines, the first article test shall verify the lateral critical speeds of the gas turbine engine. Actual torsional critical speeds below the trip speeds shall be confirmed by test stand data.

a. If the critical speed(s) of rotors with hydrodynamic bearings encroach upon the specified operating speed range, the rotor sensitivity to the vibration shall be demonstrated.

b. For anti-friction bearing designs, it is recognized that acceptable critical responses may occur in the operating speed range. These responses shall be identified prior to testing.

4.6.4.1.2 Gas turbine engine bleed air demonstration. Gas turbine engine first article inspection testing shall validate cycle deck predicted customer bleed available from idle to full power for ambient temperatures from -29 to +49 °C (-20 to +120 °F).

4.6.4.1.3 Gas turbine engine performance. Throughout the first article inspection testing, gas turbine performance shall be monitored, including power output, fuel consumption, and aerodynamic temperatures and pressures as specified (see 6.2).

4.6.4.1.4 Engine instrumentation and protective devices. The proposed engine instrumentation and protective devices, as a minimum, shall be installed for control and protection of the gas turbine engine during endurance testing. Engine instrumentation shall be sufficient to measure power output, fuel consumption, vibration, and engine efficiencies across the power band.

4.6.4.1.5 Lubricating oil consumption and analysis.

4.6.4.1.5.1 Lubricating oil consumption. Lubricating oil consumption shall be measured throughout the standard endurance test. All additions or removals of lubricating oil from the engine shall be measured. Lubricating oil consumption rates shall be determined at approximately each 250 hours of testing or more frequently as recommended by the manufacturer. Oil consumption shall not exceed the limit as established by the manufacturer prior to commencing the test.

4.6.4.1.5.2 Lubricating oil analysis. Each lubricating oil sample shall be analyzed in accordance with the testing requirements of MIL-PRF-23699. In the case of industrial gas turbine designs using MIL-PRF-17331 (2190 TEP) oil, standard analysis procedures from that specification shall be used. Two lubricating oil samples shall be taken from each lubricating oil circulating system at the following times during the endurance test:

a. Prior to the start of the testing.
b. During the first cycle.
c. During the last cycle.
d. After approximately each 250 hours of testing.
e. During the cycle preceding and after every lubricating oil change.

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4.6.4.1.5.3 Lubrication oil analysis conduct and criteria. Lubricating oil shall be analyzed in accordance with the methodologies and criteria of the Joint Oil Analysis Program (JOAP) in accordance with NAVAIR 17-15-50.1, .2, .3, and .4.

4.6.4.1.6 Fuel oil analysis. Two samples of fuel used for test engine operation shall be collected at initial fueling and then at 250-hour intervals during the endurance tests. Samples shall be labeled indicating test cycle number and endurance test hours. Samples shall be analyzed in accordance with MIL-DTL-16884 requirements, or, if an alternative fuel is allowed, in accordance with the parameters stated in table VII using the test procedures of MIL-DTL-16884.

4.6.4.1.7 Intercooling/injection cooling water analysis. Samples of cooling water shall be collected from the cooling system, if applicable, at the beginning and end of the endurance test and at intervals of 250 hours of endurance testing. Samples shall be labeled. Each sample shall be analyzed to determine inhibitor concentration.

4.6.4.1.8 Exhaust gas emissions. Gaseous and particulate exhaust gas emissions shall be measured utilizing the methods outlined in 4.6.9. Gaseous and particulate exhaust emissions shall be measured at the beginning and at the end of the standard endurance test and shall not exceed those stated in table XII. Smoke emissions shall be measured during the first and last standard endurance test cycle. The opacity at any of the measurement points shall be in accordance with 4.7.5.

4.6.4.2 Maintenance during endurance testing. From the start of test, no parts shall be modified, replaced, or adjusted, except for pre-approved scheduled maintenance and to correct permissible shutdowns as defined in 4.6.4.3.

4.6.4.3 Shutdowns during endurance testing. Engine shutdowns shall be defined as either permissible shutdowns or forced shutdowns. As specified (see 6.2), all alarm conditions and forced shutdowns during testing shall be captured, including time stamps and corrective action. In addition, the engine shall be shut down if one or more of the following conditions occur:

a. Abnormal operation or failure of an engine component.
b. Engine operation outside of operating parameters.
c. Engine torque or speed exceeding acceptable range of variability.
d. Lubricating oil exceeding any physical test condemning limit.
e. Test personnel or engine endangered.
f. Faulty test cell service systems or other systems extraneous to engine.

4.6.4.3.1 Permissible shutdowns. Permissible shutdowns include the required shutdown period at the end of each test cycle, shutdowns due to facility problems, and shutdowns at the end of a test day. Scheduled maintenance shall only be performed during these shutdown periods as necessary.

4.6.4.3.2 Forced shutdowns. An engine shutdown shall be classified as a forced shutdown if it can be described by one or more of the following:

a. A single failure of an engine external component that cannot be corrected or repaired within a 2-hour period (if the engine shutdown occurred during off-hours, initiation of the repair may be delayed until the start of the next testing day). The 2-hour period does not include obtaining replacement parts, tools, or personnel to perform the repair.
b. Two failures of the same external engine component.
c. An engine component repair, replacement, or adjustment during the test that requires access to the internal engine components (i.e., parts requiring engine disassembly).
d. Failure to maintain engine operation within the variability limits of the operating parameters, target torque, or target speed or failure to maintain lubricating oil within the condemning limits.
e. Any failure that causes cessation of the test.
f. If the forced shutdown falls within category b, or e, then the test shall be ruled a test failure.
g. If the shutdown is a category a, c, or d, a solution to the failure shall be provided to correct the problem; otherwise, the test shall be ruled a test failure.
h. If a forced shutdown is ruled a test failure, then the endurance test shall be terminated. Recommencement of the entire endurance test at time zero shall then be required.

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4.6.5 Post-endurance testing examinations. Engines and enclosure shall be examined at the conclusion of endurance testing in accordance with 4.6.5.1 through 4.6.5.2.

4.6.5.1 Post endurance test evaluation. Upon completion of first article endurance testing, the engine clearances and wear shall be measured, including those items in table V. The engine and its attached and unattached assemblies shall be disassembled. Each piece shall be re-inspected, and measurements shall be made before commencing any other testing. Special tools furnished with the engine shall be used during this inspection.

4.6.5.1.1 Evaluation criteria. The failure of any component to meet the manufacturer's wear limits or the evidence of cracks, cavitation, pitting, scuffing, galling, fretting, or overheating shall be cause for the rejection of the engine. A failure to meet evaluation criteria shall be cause for the rejection of the engine. The engine's failure to meet requirements shall result in a restart of the first article tests.

4.6.5.1.2 Reassembly. After the inspection is completed, the engine shall be reassembled to overhaul standards.

4.6.5.1.3 Reassembly testing. The reassembled unit shall successfully complete one complete cycle of the applicable standard endurance test.

4.6.5.2 Post-test borescope. Following testing completion, a full borescope inspection of the engine shall be completed.

4.6.6 Endurance testing acceptance criteria. The endurance test profile shall be considered complete after the engine has successfully run all the requirements of the testing and the following conditions are met:

a. No shutdown or fail to start has occurred because of engine malfunction.

b. No part failure has occurred that cannot be corrected within the 1-hour shutdown time between cycles using only onboard repair parts and consumables. In any event, two malfunctions or part failures involving the same part or component shall be cause for test failure. Any malfunction or part failure requiring access to internal engine components shall be cause for test failure.

c. No internal component of the engine (i.e., parts requiring engine disassembly or partial disassembly) has been replaced, repaired, or adjusted during the test.

d. No external component has been replaced or repaired other than as may be required by engine malfunction b above. Preventive maintenance parts identified by the manufacturer and the Technical Authority prior to start of the test are excluded.

e. Vibration levels shall not have exceeded the manufacturer's limits for production acceptance.

f. Lubricating oil and cooling water analyses shall not have exceeded the manufacturer's limits for in-service engines.

g. Components presented for inspection shall not have exceeded the manufacturer's limits for continued service.

TABLE XI. Allowable gas turbine engine degradation.

Nomenclature	Allowable degradation, %
SFC	5
Power	0 (shall continue to make rated power)
Turbine control temperature	3

h. All gas turbine engine performance parameters are within production limits. The gas turbine engine continues to produce the rated power, and the gas turbine engine turbine temperature and specific fuel consumption (SFC) has not been degraded by more than the allowable degradation percentage subject to any maintenance schedule to be performed as established prior to commencing the first article testing. The maximum allowable degradation percentage during the test shall be as specified in table XI.

i. The SFC increase shall be the difference in the SFC at the start of the test and at the end of the test. The increase in SFC shall not exceed the percent allowable degradation for each distinct power level in the endurance

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test cycle. All test data shall be corrected to NSD conditions defined in table I, except allowance shall be made for lower inlet and exhaust losses at power levels less than 100 percent.

j. The cumulative time due to unplanned engine shutdowns that are corrected or repaired within 2 hours shall not exceed 1 percent of the test length. The test may be continued as needed to achieve a period of testing equal to the required test length that meets the requirements. The shutdown period following each cycle, shutdowns due to facility problems, and shutdowns at the end of the test day do not count in determining the total unplanned engine shutdown time.

k. All enclosure requirements (if enclosure was included in endurance testing) were met throughout the duration of the testing.

4.6.7 Shock testing. When required (see 6.2), the engine shall be tested for grade A shock resistance in accordance with MIL-DTL-901. The complete engine, including the engine, marine gear or generator (if attached directly to the engine), instruments, controls, attached and detached accessories, and shipboard interfaces, shall be tested as a principal unit. When enclosures are included, they shall be included in the shock testing. Where resilient mounts are used to isolate the sub-base or enclosure from the hull structure, the shock requirements shall apply to the unit as installed on the resilient mounts. Shock testing shall be conducted after a visual and dimensional examination of 4.5.

4.6.7.1 Examinations after shock test.

4.6.7.1.1 Preliminary evaluation. After completion of the shock tests and before removal from the shock platform, the engine and its accessories shall be inspected as follows:

a. The complete unit shall be visually examined for breakage, distortion, cracks, leaks, and relative movement of components, housing, major assemblies, and accessories.

b. Inspection covers shall be removed and accessible internal areas visually examined for damage. A borescope evaluation shall be conducted.

c. All lubricating oil filters and strainers shall be inspected.

4.6.7.1.2 Detailed examination. The engine unit shall be returned to the gas turbine engine manufacturer or agreed upon facility for detailed examinations and operational tests as specified herein. Upon receipt at the final inspection site and before removal from the transport, the unit shall be given a visual examination to ensure that no damage occurred during transit.

a. The unit shall be disassembled, and bearings and moving parts shall be examined for excessive wear or damage attributed to the shock and post-shock tests. The internal surfaces of the block, housings, accessories, and mounting hardware shall be examined for evidence of shifting, rubbing, momentary contact of moving parts, and cracks.

b. The complete unit shall be visually examined for rotor-to-casing rubs, breakage, distortion, cracks, leaks, and relative movement of components, housings, major assemblies, and accessories.

4.6.7.2 Evaluation criteria. The failure of any component to meet the manufacturer's wear limits shall be cause for rejecting the engine. The unit shall exhibit none of the impermissible damage specified in 4.6.7.2.1.

4.6.7.2.1 Impermissible damage. Damage as characterized or as specified as follows shall not be permitted as a result of shock test:

a. Damage that will result in failure to maintain engine operation within the limits of the manufacturer's operating parameters, rated power, or rated speed or failure to maintain lubricating oil from exceeding the manufacturer's condemning limits.

b. Damage that renders inoperative any item or device provided to protect the equipment or operating personnel, such as speed-limiting governors, overspeed trips, or shutdown devices.

c. Shock-induced actuation of safety devices specified in item b above.

d. Damage or mis-adjustment of speed-governing systems that results in inability to maintain speed automatically within the limits imposed by the individual equipment specifications until repairs or adjustments are made.

e. Loosening or shifting of any item affixed to or housing a rotating part.

f. Damage that causes any assembly or subassembly to detach from any of its intended restraints.

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g. Damage that requires repairs of any type to be made before equipment operating parameters can be adjusted to suit operational requirements.

h. Damage that creates a safety hazard to personnel or equipment.

i. Fracture of any structural part or supporting member.

j. Stripping or failure by gross distortion of engaged threads (male or female) or any threaded part or fracture (cracking) of any fastener.

k. Increase of dynamic imbalance of any rotating part or assembly to such an extent that the affected item requires rebalancing to meet the specified balance tolerance.

l. Damage causing misalignment requiring correction before the engine can be operated at rated speed and power.

4.6.7.2.2 Permissible damage. Damage from shock tests as characterized or specified as follows shall be permitted to the extent or under the conditions listed for each item:

a. Damage that shortens the useful life of wearing parts by initiating or aggravating some long-term detrimental effect such as wear, fatigue, or erosion is permissible if the useful life of the wearing parts is reduced by no more than 25 percent.

b. Damage resulting in minor leaks of fluids or gases insofar as no requirement of this specification is violated by any consequences of the leakage.

c. Loosening of hold-down devices if no requirement of this specification is violated as a consequence of such loosening.

d. Damage that requires adjustment to control or equipment set points to regain proper operation or damage that requires repairs or compensatory action to prevent further (and eventually unacceptable) damage that is related to increased wear rates or other long-term effects.

e. Damage that brings about (until repairs are made) a requirement for slight-to-moderate increase of manual effort, maintenance, or attention on the part of equipment operators to maintain the operating mode or the machinery space environment.

4.6.7.3 Post-shock test evaluation re-testing.

4.6.7.3.1 Reassembly. After the detailed examination is completed, the engine shall be reassembled and subjected to the post-shock tests specified in 4.6.7.1. The attached and detached accessories and enclosure (if applicable) that were previously with the unit during the shock test shall remain with the unit for the post-shock test re-testing.

4.6.7.3.2 Re-testing. If the post-shock test examination reveals no impermissible damage as specified in 4.6.7.2.1, the unit shall be subjected to post-shock test testing consisting of four cycles of the appropriate type from tables VIII or IX, and each cycle shall be followed by a 10-minute minimum duration shutdown period.

4.6.7.4 Shock test acceptance criteria. Completion of the re-testing of 4.6.7.3.2 and the acceptance criteria of 4.6.6 shall constitute successful shock testing.

4.6.8 Inclined operation test. When required (see 6.2) and a base and enclosure is provided, an inclination test or analysis shall be performed. The inclination test on the completely assembled unit shall be made under the conditions of time at the load specified in 4.6.8.1 in each of the following positions:

a. Shaft inclined, forward end down, 5 degrees.

b. Shaft inclined, forward end up, 5 degrees.

c. Shaft horizontal, base inclined to the right 15 degrees.

d. Shaft horizontal, base inclined to the left 15 degrees.

4.6.8.1 Inclined operation test conduct. Prior to start of tests, the lubricating oil sump shall be filled to the maximum operating level with the unit in a level position. No oil shall be added during the tests, and the level shall be checked and recorded prior to start of test and between each inclination change. The engine shall be run at engine speeds in accordance with the following powers:

a. 5 minutes at no load.

b. 5 minutes at 50 percent load.

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c. 5 minutes at rated load.

d. 5 intermittent load cycles, each cycle consisting of rated load for 30 seconds and no load for 30 seconds.

4.6.8.2 Inclined operation test evaluation. During the progress of the inclination test, it shall be ascertained that there has been no change in gas turbine engine mechanical vibration from the mechanical vibration encountered in the no-incline horizontal condition, that there is no pounding or grinding of bearing or rotors to stators, and that there is no oil leakage.

4.6.8.3 Inclined operation test waiver. If the gas turbine engine is an aero-derivative gas turbine, a waiver of the inclined operation test may be requested subject to approval by the Technical Authority (see 6.2). Operational demonstration of the inclined values identified in paragraph 4.6.8 or greater for flight applications of the proposed engine meet this requirement.

4.6.9 Gas turbine engine emissions/smoke test. Exhaust emission testing shall be conducted, and exhaust emissions shall be recorded for the exhaust emission constituents of NO_x, HC, CO, CO_2, O_2, and PM. Results shall be measured in specific emissions (grams per kilowatt-hour [g/kW hr]) and as follows:

a. The cycle(s) used for the test(s) shall conform to the engine type.

b. Gaseous emissions shall be measured and confirmed at the beginning and at the end of the standard endurance test. Particulate emissions shall be measured at the end of that test.

c. Exhaust smoke emissions shall be measured in accordance with the method used in ISO 11042.

d. Tested pollutants shall not exceed the limits in table XII.

TABLE XII. Emission pollutant standards for Navy marine gas turbines.

Pollutant	Limits (g/kW-hr)
NO_x	-
HC + NO_x	7.2
PM	0.2
CO	2.0
Smoke	20% opacity

4.6.10 Health hazard assessment. A health hazard assessment in accordance with MIL-STD-882 task 207 shall be conducted to ensure conformance to the requirements of 3.2.4.

4.6.11 Maintainability. A maintainability analysis shall be performed. The techniques of MIL-HDBK-470 and MIL-HDBK-472 may be utilized. The reliability and other requirements of 3.4.1 through 3.4.1.3 shall be demonstrated or proven by analyses.

4.6.12 EMI. When required (see 6.2) and a gas turbine base and enclosure is provided, the emission and susceptibility tests specified in MIL-STD-461 for ships equipment below decks shall be performed to determine electromagnetic compliance with 3.4.9.1.

4.6.13 Gas turbine inlet distortion test. The gas turbine inlet plenum shall be tested in accordance with, and comply with the requirements of, ASME MFC-26 and SAE ARP1420.

4.6.14 Airborne noise. If a gas turbine base and enclosure is provided, the airborne noise shall be tested in accordance with MIL-STD-1474.

4.6.15 Structureborne noise. If a base and enclosure is provided, the structureborne noise shall be tested in accordance with MIL-STD-740-2.

4.6.16 Gas turbine engine generator set test (previously accepted gas turbine). As specified (see 6.2), when a previously Navy-accepted mechanical drive gas turbine engine is proposed for a new generator set application and does not require any first article performance testing, as indicated by a design assessment in accordance with appendix A, the first such generator set shall undergo verification testing in accordance with MIL-G-22077.

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4.6.17 Vibration test. The engine shall be instrumented and tested to verify the previously established acceptable vibration levels in accordance with 3.4.3.14. Sufficient vibratory instrumentation shall be installed to validate the rotor dynamic models of the gas turbine engine. When a complete engine package in an enclosure is furnished, that shall be tested for compliance with MIL-STD-167-1 for type I vibration.

4.7 Other examination and tests. The following examinations and tests are applicable for first article and conformance testing in accordance with table IV, and as noted.

4.7.1 Piping leak test. Piping systems shall be tested for leakage, rupture, or deformation by hydrostatic test at 1.5 times operating pressure. No leakage is permitted.

4.7.2 Gas turbine start and start system test. Normal, cold, and emergency start tests shall be performed, as applicable, as follows in 4.7.2.1 through 4.7.2.3.

4.7.2.1 Normal starting. The engine shall be subjected to 10 consecutive starts by the start method and apparatus intended for end use. The time to start, time to reach minimum idle speed, minimum starting rpm, and minimum starter requirements shall be determined during these start tests. The time to start shall be the time required from initial engagement of the starter mechanism (the engine shall be at a standstill) to the time at which idle speed is attained. The normal starting source energy requirements shall be validated during these start tests.

4.7.2.2 Cold starting. When required (see 6.2), an additional cold start test shall be performed utilizing the intended gas turbine engine start system. For each start, the engine and batteries (if applicable) shall be placed in an atmosphere of -6.8 °C (20 °F) and cold soaked for a period of 10 hours. A rerun of the test shall be required if any of the tests is a start overtemperature shutdown, a hung start, or an aborted start for any other reason. The time to start, time to reach minimum idle speed, and minimum starting rpm shall be determined during these start tests. Critical lubricating oil pressures shall be monitored during the cold start test. The time to start shall be the time from initial actuation of the starting mechanism with the engine at a standstill to idle. The cold starting source energy requirements shall be validated during these start tests.

4.7.2.3 Emergency starting. When an emergency starting capability is required (see 6.2), a series of four emergency start sequences shall be performed with the proposed emergency start system.

4.7.3 Safety/shutdown test. All automatic shutdowns, limits, warning devices, and any other gas turbine engine or personnel protection features shall be tested in all normal and emergency operating modes and configurations. Failure to actuate the protection feature shall be cause for test failure.

4.7.4 Overspeed operation test. The gas turbine engine shall demonstrate the ability to operate up to the overspeed shutdown point without damage. The gas turbine engine and controller combination shall demonstrate the ability to secure the gas turbine engine when the overspeed trip point is exceeded.

a. A single shaft gas turbine shall be operated for 10 minutes at no load at 105 percent of NSD rating operating output speed.

b. A free power turbine gas turbine shall be operated for 10 minutes at 110 percent of NSD rating operating output speed.

4.7.5 Smoke opacity requirements. When required (see 6.2) the combustor shall not emit visible exhaust smoke except during start-up condition. The opacity at any of the measurement points shall not be greater than 20 percent.

4.7.6 Gas turbine engine enclosure exterior surface temperature. As specified (see 6.2), if a gas turbine base and enclosure is provided, the external surface temperature shall be measured during operational testing and shall not exceed 51 °C (125 °F) or shall have insulative lagging that maintains that temperature or less.

4.7.7 Air tightness. If a gas turbine base and enclosure is provided, the base and enclosure air tightness shall be determined during noise testing and during demonstration of the firefighting system. The enclosure shall demonstrate the ability to retain the firefighting agent in the correct concentration for the correct amount of time.

4.7.8 Firefighting. If a gas turbine base and enclosure is provided, the base and enclosure firefighting system shall be demonstrated to meet the requirements of 3.7.22.

4.7.9 Operational test. Production gas turbine engines operational conformance testing shall be as follows:

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a. One standard endurance test cycle as specified in tables VIII or IX as applicable to the gas turbine engine class, unless otherwise specified (see 6.2). The environmental conditions required for the first article endurance test are not required for this test.

(1) Measurements of observed load shall be corrected by ISO 2314, or a Performance Cycle Deck Model, to the standard conditions given in table I. When available, manufacturers' validated computer performance decks shall be used.

(2) Monitored parameters shall include those items specified in 3.4.13 through 3.4.13.8.

(3) Lubricating oil samples shall be collected before and after the cycle and analyzed as specified herein.

b. Starting tests shall be performed.

c. The emergency shutdown sensor outputs shall be validated.

d. Engine fuel control and speed governing shall be demonstrated during the package conformance test. Engine senor outputs shall be validated during an engine only conformance test.

4.7.10 Borescope examination. A borescope examination shall be conducted after testing in accordance with S9086-HC-STM-010/234, NSTM Chapter 234, and approved by the Technical Authority.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gas turbine engines covered by this specification are intended for use in marine applications for propulsion, generator drive, and auxiliary power. The engines are intended for use in Naval surface ships. The engines will be used for both continuous and non-continuous duty. The gas turbine engines will vary in loading from high time at low load to high time at high load. Current stock of gas turbines , enclosures, and their components and parts may be used until depleted; future requisitions for items covered in this specification, except those for spares for legacy installations, should refer to the current version of this specification (see 6.7).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of engine (see 1.2) being procured.

c. When first article inspection is required and endurance testing duration, if not as specified herein (see 3.1 and 4.6.4).

d. Rating of engine (see 3.1, 3.1.1, and 3.1.1.1) being procured.

e. When a design assessment process is requested (see 3.1, 4.4, and A.1.1).

f. SFC (see 3.1.2).

g. Mean time between failures, if other than as specified herein (see 3.4.1.3).

h. Emergency and cold start requirements, if any (see 3.4.8, 3.4.13.17, 4.7.2.2, and 4.7.2.3).

i. Start air source (see 3.4.8.2.1).

j. Inclined operation test waiver (see 3.4.11.4 and 4.6.8.3).

k. EMI testing (see 3.4.11.4 and 4.6.12).

l. Requirements for the submittal of manufacturer-defined cooling water requirements for power augmentation, emissions reduction, lubrication oil cooling, and water washing for Technical Authority review (see 3.4.11.7).

m. Ships electrical power to be supplied (see 3.4.13.9).

n. ISO 20816-3 vibration standards modification, if necessary (see 3.4.13.10).

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o. Condition monitoring and health assessment system (see 3.4.14).
p. Smoke requirements (see 3.4.16 and 4.7.5).
q. Enclosure (module) to be provided or not (see 3.7 and 4.7.6).
r. Subbase to be provided or not (see 3.7.6).
s. Firefighting agent to be used if required (see 3.7.22).
t. Fire detection control unit (see 3.7.22.4).
u. Rails, special tooling, or other apparatus to be provided (see 3.7.23.2.b).
v. Airborne noise requirements, if not as specified herein (see 3.7.24.1).
w. Structureborne noise requirements, if not as specified herein (see 3.7.24.2).
x. Shock requirements, if not as specified herein (see 3.8 and 4.6.7).
y. Testing when a gas turbine engine previously used for mechanical drive is to be used as electrical generator (see 3.9 and 4.6.16).
z. Redesign or modification information (see 4.6.1.4).
aa. Performance monitoring (see 4.6.4.1.3).
bb. Shutdown information during first article testing (see 4.6.4.3).
cc. Conformance testing operational cycle if not a specified herein (see 4.7.9.a).
dd. Conformance alarm testing if not as specified herein (see 4.7.9.e).
ee. Packaging requirements (see 5.1).
ff. Recommended documentation (see 6.3).
gg. Design assessment process report (see 6.4).

6.3 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition (see 6.2) of any gas turbine under this specification:

a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual in accordance with MIL-DTL-24784.
e. Visual aids and maintenance cards in accordance with MIL-STD-3034.
f. Test and examination reports, particularly all first article tests and examinations.
g. Certifications.
h. Provisioning technical documentation (PTD).
i. Maintenance and repair parts listings.

6.4 Design assessment process report. The assessments conducted in accordance with the DAP of appendix A should be reported in detail (see 6.2).

6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Definitions. General gas turbine engine terminology is defined in API STD 616, ISO 11086, and ISO 3977. For the purposes of this specification, the definitions of 6.6.1 through 6.6.11 apply.

6.6.1 Depot-level maintenance (D-level). Tasks requiring a higher skill, capability, or capacity than I-level maintenance. D-level maintenance tasks are focused on repair, fabrication, manufacture, assembly, overhaul, modification, refurbishment, rebuilding, test, analysis, design, upgrade, painting, assemblies, subassemblies, software, components, or end items that require specialized facilities, tooling, support equipment, personnel with higher technical skill, or processes beyond the scope of the Intermediate Maintenance Activity (see MIL-STD-3034). Also, see the definitions for I-level and O-level maintenance for comparison. For gas turbines, D-level maintenance is normally accomplished at the engine manufacturer's plant or at a commercial or military overhaul activity equipped with the recommended shop tools and special tools and facilities. This maintenance

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typically includes complete disassembly, examination, cleaning, and repair of the engine and accessories and a qualifying engine test run after overhaul.

6.6.2 Gas turbine Qualification Letter. Upon completion of the technical review of the design and construction of the engine and successful testing in accordance with the requirements herein, the Technical Authority will issue a Gas Turbine Qualification Letter. The technical review is completed when all aspects of the Design Assessment Process review, testing, and inspection are resolved to the satisfaction of the Technical Authority. NOTE: In this context, "Qualification" refers to a formal evaluation and acceptance of a gas turbine for service; this usage is not related to the Defense Standardization Program Office SD-6 qualification of products.

6.6.3 Gas turbine base and enclosure. The gas turbine base is what the gas turbine is mounted to. The gas turbine enclosure is the environmental structure around the gas turbine. For generator sets, the gas turbine base and enclosure may also have a gear, a generator, and sometimes a flywheel mounted to them. The combination of the gas turbine base and the gas turbine enclosure is sometimes known as the gas turbine module.

6.6.4 Hot section repair interval (HSRI). HSRI refers to the Time at Rated Power (TARP) of the hot section life of high pressure and low pressure turbine blades and vanes. It typically is a measure of the turbine airfoil creep life at rated power and their hot corrosion life and oxidation life.

6.6.5 Intermediate level (I-level). Tasks requiring a higher skill, capability, or capacity than O-level maintenance. I-level maintenance is normally accomplished by centralized repair facility personnel such as the Navy Fleet Maintenance Activity (FMA), submarine refit and support facilities, Regional Maintenance Centers (RMC), and Battle Group or other Intermediate Maintenance Activities (IMA) (see MIL-STD-3034). For gas turbines, I-level maintenance consists of calibration; repair or replacement of damaged or unserviceable parts, internal engine components, or assemblies; emergency manufacture of unavailable parts; and technical assistance. I-level internal maintenance is primarily accomplished in place onboard the ship.

6.6.6 Mean installed engine life (MIEL). MIEL is the mean installed life of a gas turbine engine prior to removal of the gas turbine engine from the vessel for any self-induced cause.

6.6.7 Module. The gas turbine engine is enclosed in a base and enclosure that can be referred to as the module.

6.6.8 Normal operating speed range. Speed range from idle speed through maximum engine running speed.

6.6.9 Operating speed range. Speed range from zero through engine speed limiting running speed.

6.6.10 Organizational level maintenance (O-level). O-level maintenance is the lowest maintenance echelon and consists of all maintenance actions within the capability and resources provided to the organization who routinely oversees equipment operation such as ship's force (see MIL-STD-3034). Also see the definitions for D-level (see 6.6.1) and I-level maintenance for comparison. For gas turbines, O-level maintenance normally includes overall examination of the complete engine, both external and internal, using borescope techniques plus replacement of any external components (e.g., igniters, thermocouple harness, fuel control, governor, starting systems).

6.6.11 Significant changes. Significant changes are modifications made to the engine components or systems listed in 3.4 or changes that affect engine performance, shaft dynamics, and temperatures. The modifications may be major design changes or minor modifications to improve manufacturing processes or service life. Any engine component modified such that it is no longer interchangeable with the component it replaces is considered to have undergone a significant change.

6.7 Supersession data. This specification also supersedes American Bureau of Shipping Naval Vessel Rules (ABS NVR) 2-3-1 and Naval Combatant Design Specification (NCDS) Addendum 234-A. Existing designs that have a significant change (see 6.6.11) to any part should follow the design assessment process specified herein to evaluate the acceptability of the new part and to determine what verification or testing specified herein, if any, should be conducted to verify the acceptability of the part.

6.8 Subject term (key word) listing.

Bleed air
Combustor
Compressor
Flat rating

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Generator set
Intercooler
Navy Standard Day (NSD)
Overspeed
Power turbine
Recuperator
Regenerative
Tent rating
Vibration

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

MARINE GAS TURBINE DESIGN ASSESSMENT PROCESS (DAP)

A.1 SCOPE

A.1.1 Scope. This appendix covers the design assessment process (DAP). When requested (see 6.2), a DAP is intended to be used to determine if the design and maturity warrants a reduction in required first article endurance testing, or adjustment to other testing requirements, or that an existing gas turbine of a particular navy type may be accepted for use as a different type, or that uprating is appropriate. This appendix is a mandatory part of the specification when invoked (see 3.1 and 6.2). The information contained herein is intended for compliance.

A.1.2 Applicability. Gas turbine engines for naval shipboard usage may require an assessment of the design and first article testing and acceptance, or uprating. The DAP assesses the engine service history, the data required by this specification, and other data requested to support the DAP; the DAP then determines the potential naval marine rating, the potential uprating of existing engines, and which service category applies.

A.1.2.1 DAP process. The DAP assesses:

a. The proposed gas turbine engine overall design.

b. The proposed gas turbine engine projected life with the proposed rating or uprating.

c. The projected reliability of the proposed gas turbine engine for marine naval service at that rating.

d. The long term producibility and supportability of the proposed gas turbine engine design in order to determine the proposed gas turbine engine service history category as a basis for adjustments to first article testing.

A.1.3 Gas turbine engine service history categories. The proposed gas turbine engine service history category determination is to be based on the gas turbine engine service history as determined by the DAP. Gas turbine engines that meet the requirements of categories II, III, or IV are eligible for reduced first article testing as described herein. The engine service history categories consist of the four categories of A.1.3.1 through A.1.3.4.

A.1.3.1 Category I. Category I is a research and development gas turbine engine that has no historical commercial or military production or service history in the proposed configuration. This includes a gas turbine engine that is configured with different engine model modules from different parent engines (for example, low pressure compressors and turbines, high pressure compressors and turbines, and combustors combined from different parent engines). Gas turbine engines under this category undergo testing as delineated in this specification.

A.1.3.2 Category II. Category II is a production derivative gas turbine engine that is a derivative of a non-marine production gas turbine engine with validated design changes for marine service application at the specified rating and required service life. The parent gas turbine engine is in production, has been certified by a certifying agency, and is a stable design under configuration control.

A.1.3.3 Category III. Category III is a production gas turbine engine with a marine service record and certified by a commercial marine certification agency such as ABS, Bureau Veritas (BV), Det NorskeVeritas (DNV), Lloyd's of London, Registro Italiano Navale (RINA S.p.A.), and others, at the proposed rating. Additional considerations are as follows:

a. The gas turbine engine population has not less than four engines in marine service.

b. These gas turbine engines have demonstrated the proposed rating, fuel efficiency, and reliability required by the proposed naval application.

c. At least one of these gas turbine engines has at least 12,500 hours of accumulated operation without overhaul, removal, or replacement of any internal or hot section parts.

d. The accumulated running time of all the gas turbine engines within the population at the proposed rating is at least 40,000 hours.

e. At least one gas turbine engine has been produced within the previous 2 years.

A.1.3.4 Category IV. Category IV is a marine gas turbine engine previously approved by the Technical Authority (see 6.5) with U.S. Navy marine service history that is being proposed for increasing the full power rating or changing service application. Additional considerations are as follows:

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a. These gas turbine engine population has not less than four engines in marine U.S. Navy service.

b. These gas turbine engines have demonstrated the proposed rating, fuel efficiency, and reliability required by its current U.S. Navy service at its current rating.

c. At least one of these gas turbine engines has at least 1,000 hours for type A engines and 12,500 hours for type B engines of accumulated operation without overhaul, removal, or replacement of any internal or hot section parts.

d. The accumulated U.S. Navy marine running time of all the gas turbine engines within the population at the proposed rating is at least 4,000 hours for type A engines and 50,000 hours for type B engines.

e. At least one gas turbine engine has been produced within the previous 2 years.

A.1.4 Conflict. If the Technical Authority does not concur with the proposed test category selection by the gas turbine engine manufacturer or contractor, the proposed gas turbine engine shall complete a category I endurance verification test in accordance with the requirements of section 4 of this specification.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in Appendix A of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Appendix A of this specification, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-G-22077 - Generator Sets, Gas Turbine, Direct and Alternating Current Naval Shipboard Use

(Copies of this document are available online at https://quicksearch.dla.mil.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN PETROLEUM INSTITUTE (API)

API STD 616 - Gas Turbines for the Petroleum, Chemical, and Gas Industry Services

(Copies of this document are available online at www.api.org.)

A.3 REQUIREMENTS

A.3.1 DAP. When required (see 6.2), the proposed gas turbine engine shall complete a DAP for naval shipboard usage in accordance with the requirements herein. The DAP of the proposed gas turbine engine shall be directed and assessed by the Technical Authority in conjunction with the gas turbine engine manufacturer or contractor.

A.3.2 Conduct of the DAP. The DAP consists of the analyses required by API STD 616, annexes B and C, and the following:

a. DAP API STD 616 analyses shall utilize Navy Standard Day (NSD) conditions (see table I).

b. DAP reliability and mean-time-between-failure (MTBF) shall be in accordance with 3.4.1.1, 3.4.1.3, and 3.4.13.2.

c. DAP shall include a demonstrated service record (DSR), consisting of operating histories of engines that are installed in different aircraft, facilities, or ships. The DSR historical data shall be detailed enough to substantiate compliance with the gas turbine engine test category recommended by the manufacturer or contractor.

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d. DAP shall include assessments of proposed engine's ability to pass required shock testing, noise testing, start testing, and other tests required by this specification (see section 4), and if modifications of those tests should be considered by the Technical Authority.

A.4 DETERMINATIONS

A.4.1 Endurance testing. When a DAP determination is made that reduced endurance testing is allowable, the endurance testing shall be in accordance with table A-I. Endurance test cycles referred to in table A-I are as described in tables A-II through A-V.

TABLE A-I. Engine rating verification performance endurance testing.

Duty type	Category I test cycle	Category II test cycle	Category III test cycle	Category IV test cycle
Type A or APU proposed for this use	1,500 hours Test cycle A	1,000 hours Test cycle A	750 hours Test cycle A	500 hours Test cycle A
Type B	3,000 hours Test cycle B	1,500 hours Test cycle B	1,000 hours Test cycle C	500 hours Test cycle D

A.4.2 Other first article testing. All other requirements of section 4 shall apply, unless adjusted due to assessment by the Technical Authority.

A.4.2.1 Clarifications. The following clarifications shall apply to the testing identified in table A-I.

a. Hours listed in table A-I test cycles are the engine run times for the required testing. These hours do not include the engine shutdown time during each cycle.

b. For a gas turbine generator application, any engine previously approved by the Technical Authority for naval mechanical drive application that does not require testing because there is no increase in rating shall undergo a generator set package test in accordance with MIL-G-22077, including those engines that exceed a generator set rating of 3,500 kW.

c. Engines to be certified for both gas turbine main (GTM) propulsion drive and gas turbine generator (GTG) AC electrical drive shall be tested for half of the test cycles with a variable power turbine output speed load curve and half of the cycles at constant output speed load curve.

d. Ratings are independent of mechanical or electrical drive applications (GTM or GTG). Each gas turbine that has successfully completed verification testing where the Technical Authority has issued a rating verification letter may be used in either application with the following exceptions:

(1) For type B electrical drive applications, where the gas turbine is mechanically coupled to a generator that in turn is electrically coupled to an electrical distribution system, the gas turbine rating being tested shall be considered equivalent to the 110 percent power rating at NSD conditions.

(2) Existing mechanical drive engines previously approved by the Technical Authority as type A low usage shall be retested as type B high usage for any use as a type B high usage generator.

(3) Existing electrical drive engines previously approved by the Technical Authority as type A low usage shall be retested as type B high usage for any use as a type B high usage generator.

(4) An engine that has been first article duty tested for a type B application shall be considered to have met the first article duty type test requirements for a type A application at the same or lower power rating.

(5) Existing engines approved as an APU that are proposed for generator applications (low or high usage) shall be tested against the requirements for generator applications (low or high usage as applicable).

A.4.3 Endurance test cycles. Endurance test cycles referred to in table A-I are as described in tables A-II through A-V.

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TABLE A-II. Endurance test cycle A.

Step	Load (percent)	Time (minutes)
1	Idle	2
2	100	10
3	20	5
4	Idle	3
5	60	10
6	100	20
7	90	10
8	0 (not operating)	60 1/

FOOTNOTE:

1/ Upon completion of step 7, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20 percent] may have to be reduced).

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TABLE A-III. Endurance test cycle B.

Step	Load (percent)	Time (minutes)
1	Idle	5
2	30	20
3	50	15
4	100	60
5	90	20
6	80	20
7	100	30
8	60	20
9	Idle	20
10	100	30
11	90	40
12	100	30
13	20	60
14	50	30
15	Idle	1
16	100	11
17	Idle	1
18	90	1
19	Idle	1
20	100	1
21	Idle	1
22	50	1
23	90	1
24	Idle	1
25	50	10
26	20	5
27	30	5
28	0 (not operating)	60 1/

FOOTNOTE:

1/ Upon completion of step 27, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20 percent] may have to be reduced).

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TABLE A-IV. Endurance test cycle C.

Step	Load (percent)	Time (minutes)
1	Idle	5
2	30	20
3	50	12
4	100	60
5	90	20
6	80	20
7	100	60
8	60	10
9	Idle	20
10	100	30
11	90	40
12	100	58
13	20	30
14	50	15
15	Idle	1
16	100	11
17	Idle	1
18	90	1
19	Idle	1
20	100	1
21	Idle	1
22	50	1
23	90	1
24	Idle	1
25	50	10
26	20	5
27	30	5
28	0 (not operating)	60 1/

FOOTNOTE:

1/ Upon completion of step 27, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20 percent) may have to be reduced]).

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TABLE A-V. Endurance test cycle D.

Step	Load (percent)	Time (minutes)
1	Idle	5
2	30	7
3	50	5
4	100	114
5	90	20
6	80	9
7	100	57
8	60	8
9	Idle	7
10	100	57
11	90	40
12	100	57
13	20	7
14	50	7
15	Idle	1
16	100	21
17	Idle	1
18	90	1
19	Idle	1
20	100	2
21	Idle	1
22	50	1
23	90	1
24	Idle	1
25	50	4
26	20	2
27	30	3
28	0 (not operating)	60 1/

FOOTNOTE:

1/ Upon completion of step 27, the engine shall be shut down for a minimum of 60 minutes before beginning the next cycle. This non-operational time shall not be included in the total endurance test time.

NOTE:

1. For constant speed testing, the speed shall be maintained constant within the capability of the test facility (e.g., when a water brake is used as the loading device, the rpm at low power points [<20 percent] may have to be reduced).

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A.4.4 SFC degradation. The maximum allowable percentage SFC degradation during the test for each engine category listed in table A-I shall be not greater than the allowable limits in table A-VI. The gas turbine engine shall still make rated power at the conclusion of any test cycle.

TABLE A-VI. Allowable gas turbine engine SFC degradation.

Engine category	Allowable SFC degradation, %
I	5
II	4
III	4
IV	3

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 2835-2025-001)

Review activity:
DLA – GS

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